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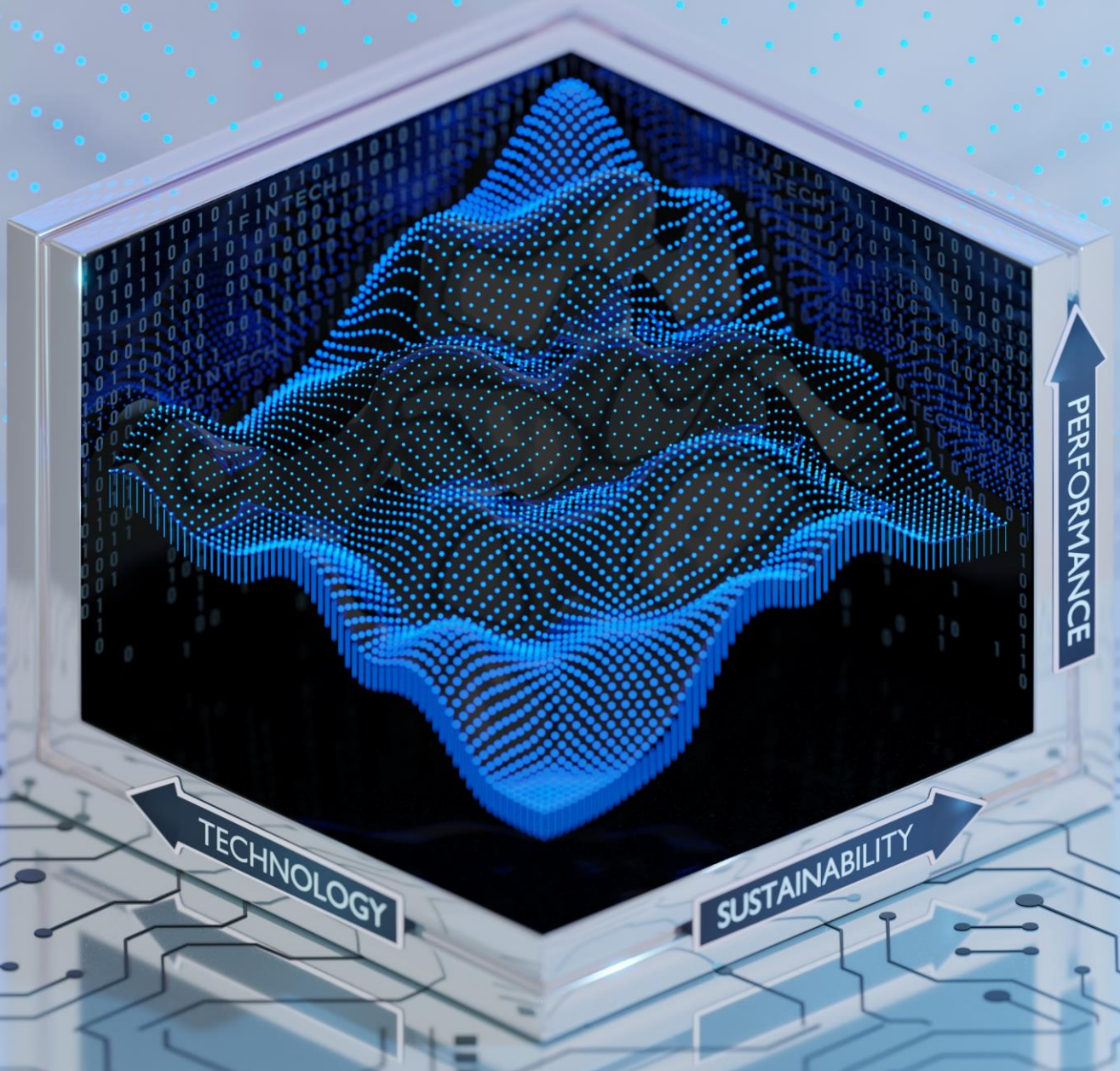

Crypto Assets Study 2022

An overview of the Swiss and Liechtenstein crypto assets ecosystem

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1. Introduction

In recent years, Switzerland and Liechtenstein have made a name for themselves as a leading centre for the development of distributed ledger technology, thus becoming known globally as "Crypto Valley". In addition to technological progress, an ecosystem for investments in crypto assets has also emerged, with various providers covering a wide range of products and services in the investment value chain.

From an investment perspective, the development of the crypto assets market can be illustrated by the total end-of-period market capitalisation shown in Figure 1.1. It shows that market capitalisation generally has risen over time with significant fluctuations between years. As per the end of April 2022, total market capitalisation amounted to over USD 1.7 billion.

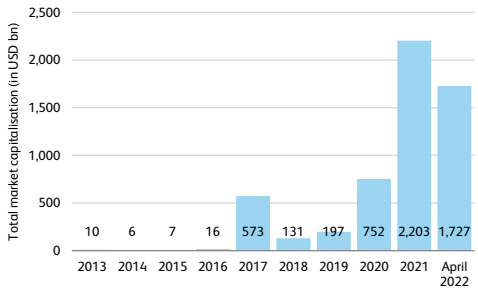

Figure 1.1: Total market capitalisation of crypto assets, by end of period (source: CoinMarketCap (2022))

In addition to the market figures, various studies show the prevalence of crypto assets among investors. A study by the Federal Reserve for the US market, for example, shows that in 2021, 12 percent of adults owned or used crypto assets in the past year, with investment purpose being the most cited reason (Lloro et al., 2022). A survey by Gemini arrives at similar figures. Their study concludes that the ownership rate of crypto assets in developed countries is around 17 percent. Globally, this rate is as high as 23 percent. Furthermore, the study confirms that investment activity in crypto assets has increased considerably in 2021 and that buying and holding crypto assets is primarily due to their long-term investment potential (Gemini, 2022). But crypto assets investments have also arrived in the institutional segment, as a study by Eurex shows. The study finds that most institutional investors which are active in the derivatives markets are invested in crypto assets or are at least interested in investing. Indirect financial products are the preferred vehicle, driven by technological and regulatory challenges for direct investments (Eurex, 2021). In general, the popularity of crypto assets among investors is likely to increase in the future (Bitstamp, 2022).

1.1. Definition of Crypto Assets

As the present study focuses on assessing the investment activities surrounding crypto assets, a definition of the subject is needed. Analogous to the last edition of this study, the term "crypto assets" is defined as follows:

> Crypto assets are digital representations, like claims, values, or rights, issued on a distributed ledger, such as a blockchain protocol, in the form of tokens.

Crypto assets are thus understood as any form of tokenised assets that are stored using distributed ledger technology. These can be, for example, pure cryptocurrencies such as Bitcoin, without any other underlying, but also tokenised derivatives on real assets such as fiat-backed stablecoins. Non-fungible tokens (NFTs) such as property rights in decentralised metaverses (e.g. LAND in Decentraland) or tokenised digital arts (e.g. Bored Ape Yacht Club tokens) also belong to the class of crypto assets.

In principle, the definition of crypto assets used in this study does not imply any requirement for their degree

of decentralisation. Tokens on public DLT infrastructures therefore qualify as crypto assets just like those on private or consortium ledgers.

1.2. Methodological Approach

The present study focuses on, firstly, giving an overview on the business models of companies active in the Swiss and Liechtenstein investment ecosystem for crypto assets. This goal is to be achieved by means of a survey of active companies incorporated in Switzerland and Liechtenstein that, according to the FinTech grid[1], offer crypto assets-related products and services in the product areas of *Deposit & Lending*, *Investment Management*, and *Banking Infrastructure*, and are based on technological concepts from the *Distributed Ledger Technology* category (see Figure 1.2).

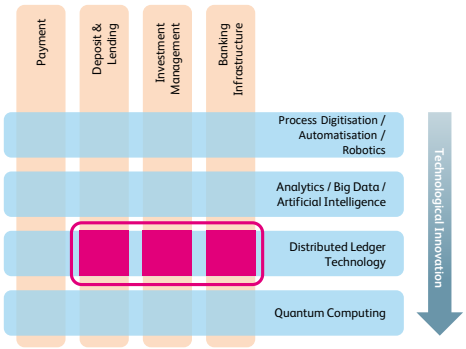

Figure 1.2: Scope of the study

The relevant companies were identified based on a proprietary database of FinTech companies enriched with established financial institutions relevant to the study and identified through continuous monitoring of the respective market.

During a period between May and June 2022, a total of 133 companies were invited to participate in the survey. 32 companies disclosed general information and key figures as of the end of 2021. The corresponding factsheets can be found in Chapter 7, reporting most of the non-confidential information collected.

The second focus of the present study is to provide, wherever possible, an estimate of the volumes of the observed business models in the Swiss and Liechtenstein investment ecosystem for crypto assets. On the one hand, data from subscription-based providers Bloomberg and Morningstar is considered. On the other hand, data provided by the two Swiss exchange operators SIX and BX Swiss as well as publicly available information is used.

1.3. Structure of the Study

After providing an introduction to the topic and its relevance in this first chapter, Chapter 2 provides a structuring framework for the crypto assets investment ecosystem. Chapter 3 provides an overview of the business models of the companies that participated in the survey, with the corresponding company factsheets listed in Chapter 7. Chapter 4 contains an analysis of the observable business volumes in the investment ecosystem for crypto assets in Switzerland and Liechtenstein and Chapter 5 analyses a crypto asset investment from a Swiss investor's perspective. Chapter 6 concludes the study.

[1] For more information on the FinTech grid, see Ankenbrand, Bieri, Frigg, Kronenberger, and Reichmuth (2022).

2. Structure of the Investment Ecosystem for Crypto Assets

Analogous to the investment value chain for traditional financial assets, there are also various processes for crypto assets that are relevant for corresponding investments. An overview of these processes and the relevant providers is given in Figure 2.1. The vertical levels define the four most important process steps required for crypto investments, i.e. issuance, investment services, trading infrastructure, and post-trading infrastructure, in addition to the level of the investor, who is the final risk bearer of a crypto assets investment and thus represents the end customer.

Besides the vertical layers, Figure 2.1 also distinguishes between three horizontal layers, defining the degree of DLT that is involved. The reason for this is that it is not only possible to invest directly in crypto assets, but there are also more and more indirect financial products such as funds or structured products available for investors. The investment value chain of these indirect financial products mainly takes place off-chain (magenta layer), i.e. without the direct involvement of DLT. Direct investments, in contrast, require the inclusion of DLT. Here, however, a distinction can be made between the degree of centralisation of the solutions or the providers. While the value chain for direct investments in crypto assets can partly be characterised by central providers (light blue layer), the trend towards decentralised finance (DeFi) has also led to the establishment of completely decentralised solutions (dark blue layer) in which software (so-called smart contracts) take over the intermediary function completely.

In the following, the services and their providers in the main processes of the value chain for crypto assets

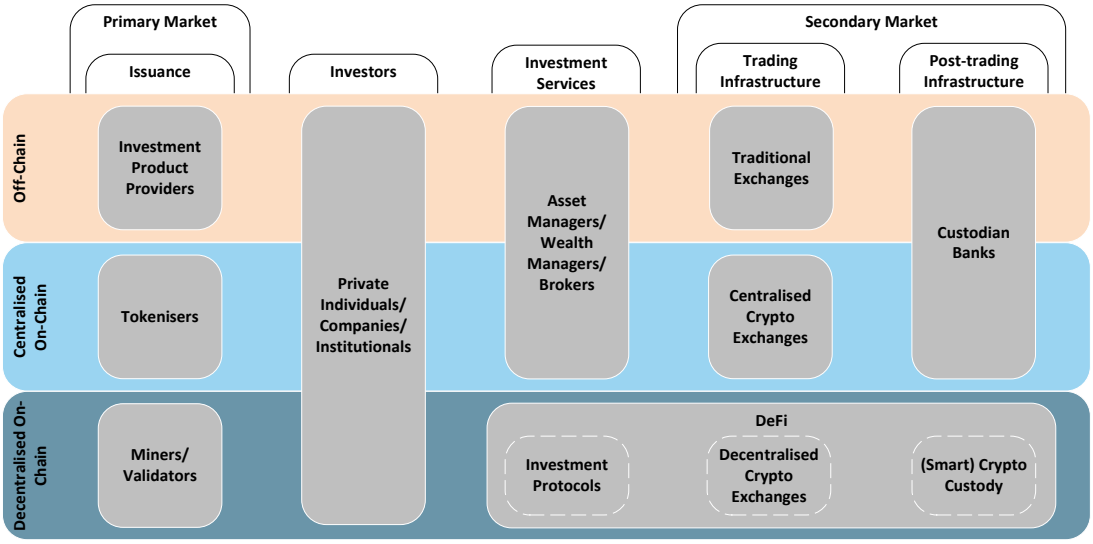

Figure 2.1: Structure of the ecosystem for crypto assets

investments, i.e. the vertical layers in Figure 2.1, are briefly discussed.[1]

Issuance

Issuers are the actors on the primary market who bring new crypto assets or indirect financial products on crypto assets to the market. The latter includes, for example, funds or structured products such as tracker certificates, which also exist for traditional assets and are therefore often issued by the same providers. For investors, these products have the advantage that they do not require any DLT-based activity, such as maintaining a personal wallet, but can be integrated directly into traditional securities accounts. Tokenisers act as issuers of new crypto assets by issuing tokens that represent an assets such as claims, values, or rights on a DLT network. These tokenisers are typically understood as a centralised provider, where exemplary tokenised crypto assets may include fiat-backed stablecoins or tokenised shares. Miners and validators can also be understood as issuers in a blockchain network, as they receive newly created crypto assets as rewards for both validating and proposing blocks to the network.

Investors

Investors represent the end customers of the value chain for crypto assets investments. With regard to business volumes, this study does not distinguish between different types of investors, like institutional or retail investors, active in the Swiss and Liechtenstein crypto assets investment ecosystem, as this is particularly difficult for direct investments due to the anonymity characteristic of DLT networks.

Investment Services

Investment services for crypto assets can be offered by asset managers, wealth managers, or brokers. On the one hand, they can include indirect investments in their offering, just as with traditional assets. On the other hand, investment services providers can also offer product and services for direct crypto assets investments. In both cases, providers act as an intermediary for the provision of investment services to their clients. However, developments in the DeFi sector have also produced corresponding decentralised solutions. These solutions include, for example, the management of outstanding credit and loan positions based on smart contracts or self-custody with a wallet.

Trading Infrastructure

Marketplaces are necessary for trading crypto asset or related financial products. Traditional exchanges can be used for indirect investment vehicles (exchange traded products or ETPs for short) on crypto assets, as these products do not involve DLT-based transactions in trading. For direct trading of crypto assets, there are centralised as well as decentralised exchanges. While centralised crypto exchanges generally work similar to traditional exchanges, i.e. by providing an order book and a corresponding matchmaking engine via an intermediary, decentralised crypto exchanges are built on top of a DLT network and operate via liquidity pools that do not require an intermediary party, except a collection of smart contracts, to orchestrate trading.

Post-trading Infrastructure

Custody of crypto assets or indirect financial products based on them represents the final process of the investment value chain. While the latter can be easily integrated into traditional security accounts of custodian banks, the custody of crypto assets requires the maintenance of a corresponding crypto wallet. The management of the wallet, or more precisely the private keys that guarantee access to the crypto assets, can again be done by a centralised provider such as a crypto bank or a centralised crypto exchange using omnibus accounts for all their clients. However, the custody of crypto assets can also be taken over by the investor in a decentralised manner by self-performing the wallet management. Furthermore, custody of crypto assets can be done in "smart" ways, such as staking or lending out assets, which in turn can generate income.

[1] For more information, see Ankenbrand, Bieri, Kronenberger, et al. (2021).

3. Overview of Providers of Crypto Assets-related Products and Services

This chapter provides an overview of the product offerings, customer segments, and key activities of Swiss and Liechtenstein companies that offer crypto assets-related investment products and services in line with the definition of crypto assets in Section 1.1 and primarily includes information provided by the companies surveyed according to Section 1.2.

In Figure 3.1, product offerings and customer segments of the companies that took part in the survey conducted for this study are illustrated. The key activities which further specify the product offering are listed in Figure 3.2. In both figures, fields highlighted in magenta mean "applicable" whereas blue fields mean "not applicable". Non-responses were also interpreted as "not applicable".

In the survey, companies were asked to classify their product offering according to the five categories *Direct investments*, *Funds*, *Structured products*, *Derivatives*, and *Infrastructure*. The latter refers to the provision of mostly IT solutions related to crypto assets investments to third parties. Due to the scope of this study, the product offering has a strong investment focus, excluding supporting services such as "know your customer" (KYC) and "anti-money laundering" (AML) identification services. Analysing the product offering and customer segments in Figure 3.1, it is evident that *Direct investments* are offered by more than half of the companies that took part in the survey, followed by *Structured products* and *Infrastructure* services. Roughly 80 percent of the companies offer only one or two of the listed products and services.

The costumers segments are displayed in the lower diagram of Figure 3.1, with the segments being divided into *Retail clients*, *Private clients*, *Corporates*, *Family offices*, *Banks*, and *Other institutionals*. In terms of their costumers segments, the majority of the companies display a diversified clientele, led by *Other institutionals*, closely followed by *Corporates*, and with *Banks* and *Family offices* in third place. *Private clients* and especially *Retail clients* are less frequently targeted customer segments, which might be partly explained

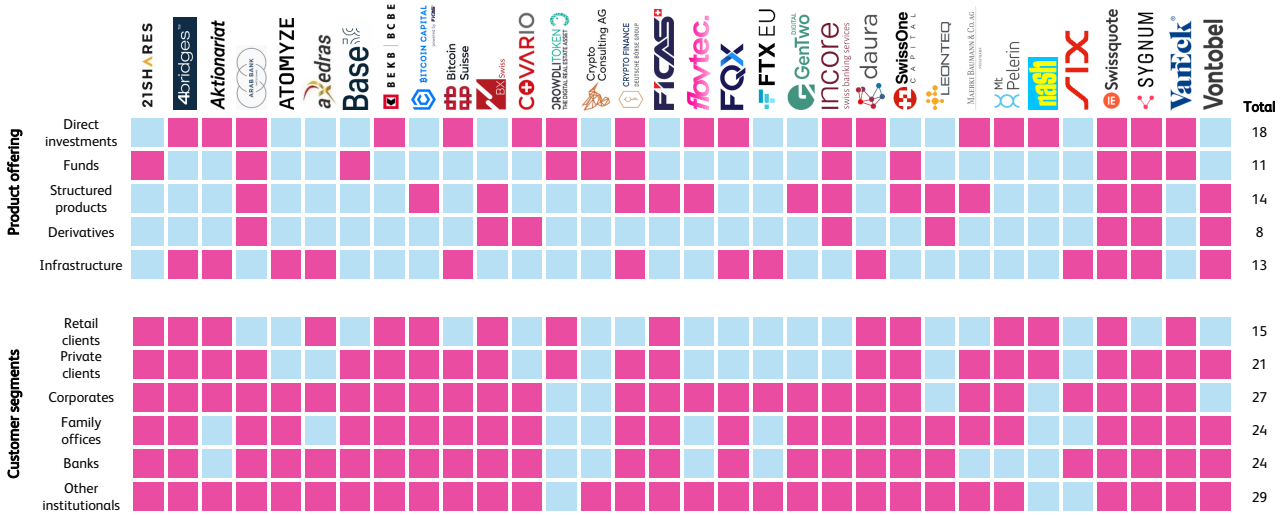

Figure 3.1: Product offerings and customer segments of companies from factsheets received

by lower investment volumes and strong international competition.

The key activities of the companies surveyed are illustrated in Figure 3.2 and include *Issuing/Tokenisation*, *Asset Management/Wealth Management*, *Brokerage*, *Exchange Services*, *Custody*, and *Lending* activities. Around two thirds of the companies pursue solely one or two of the key activities. The most often named activity is *Issuing/Tokenisation*, followed by *Asset Management/Wealth Management* and *Exchange Services*. *Brokerage* and *Custody* follow on fourth place. The least named activity is *Lending*, which seems to require a larger organisational structure and capital base or, on the contrary, a limitation to specific customer segments.

Furthermore, most of the key activities generally offer the potential for synergies with one or more other key activities. However, as most companies pursue only one or two key activities it seems that companies are highly specialised and, as hypothesised for *Lending*, the size of a company and its capital base factor in. Nevertheless, synergies between key activities might also be realised by focusing on specific customer segments.

In conclusion, Swiss and Liechtenstein companies offer diverse investment products and services for crypto assets but the majority of the companies focus on one or two product offerings and key activities. Companies with more than three product offerings and key activities often have a stronger banking background. *Direct investments* are offered more frequently than indirect investments, and in terms of the key activities, *Issuing/Tokenisation* is pursued the most. Corporate and institutional customers are the most relevant customer segments, while fewer products and services are aimed at retail customers. Overall, the market seems rather fragmented with only few companies covering the entire investment value chain.

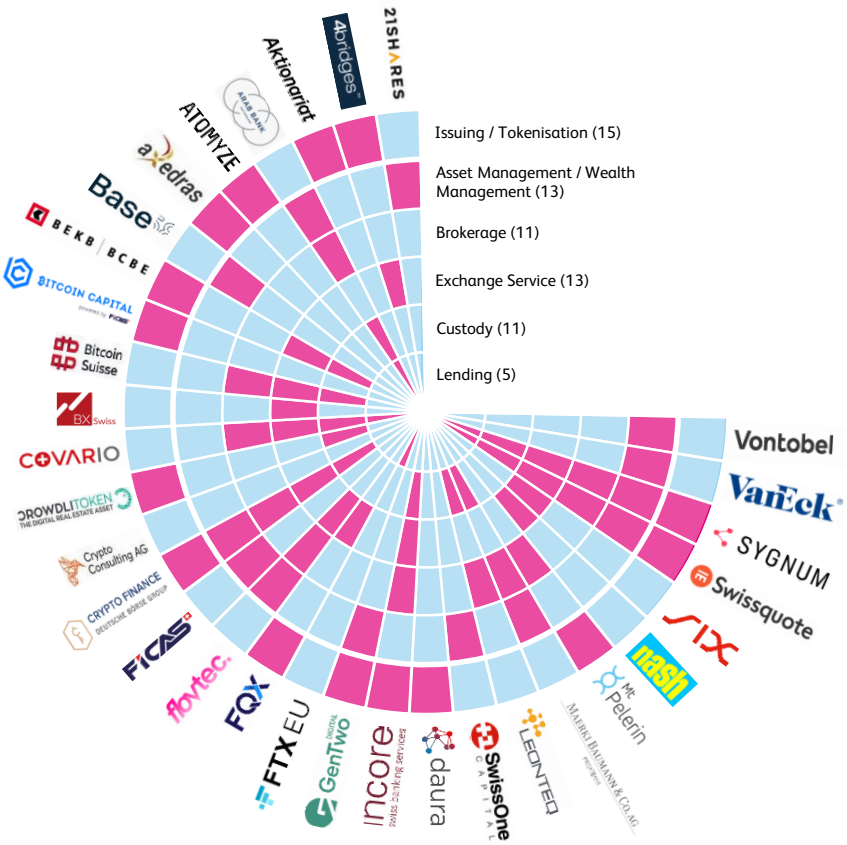

Figure 3.2: Key activities of companies from factsheets received

4. Overview of Market Activities

In this chapter, business volumes related to crypto assets investments are analysed, focusing on both indirect (Section 4.1) and direct investments (Section 4.2).

4.1. Indirect Investments

Indirect investment vehicles provide a way to gain exposure in the market for crypto assets without the need of managing corresponding tokens directly with the use of DLT. This means that certain technological necessities of direct investments, such as keeping a crypto wallet or interacting with (differently regulated) crypto exchanges, are not necessary for indirect financial products, which makes them relevant for certain types of traditional investors (Eurex, 2021).

In Switzerland, exchange traded products and open-end funds have proven to be a popular form of indirect investment in crypto assets. This is illustrated in Figure 4.1, which shows the total number of individual ISINs of ETPs and open-end funds that are either available for sale, domiciled, or traded in Switzerland and/or Liechtenstein.[1]

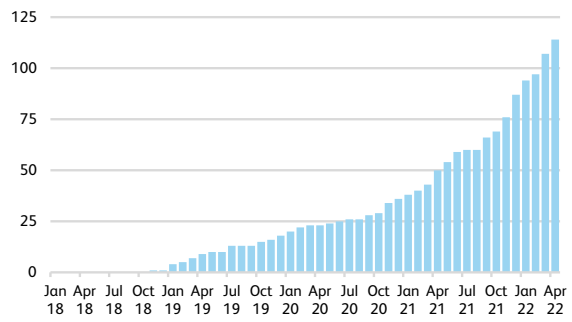

Figure 4.1: Number of individual ISINs of ETPs and open-end funds (source: Morningstar Direct)

The figure reveals that there has been a steady growth of corresponding products over the past years, starting in November 2018, the month in which the first ETP was launched. Further products were introduced over the following months. As of the end of April 2018, there were already 114 individual ISINs for ETPs and open-end funds in the Swiss and Liechtenstein crypto assets ecosystem, underlining the popularity of these indirect investment vehicles.

The development of total assets under management in these products over time is shown in Figure 4.2. The light blue curve/area shows the corresponding volume in millions of Swiss francs on the left scale. This shows that total assets began to rise sharply in October 2020 and peaked in November 2021 at a little over CHF 6.0 billion. By the end of the observation period in April 2022, the volume has fallen to around CHF 4.0 billion.

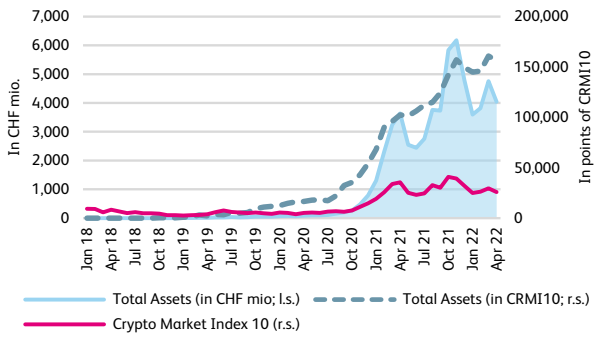

Figure 4.2: Total assets of ETPs and open-end funds (sources: Morningstar Direct, Bloomberg[2])

The development of the CHF-denominated total assets managed in ETPs and open-end funds in the Swiss and Liechtenstein crypto assets ecosystem correlates strongly with the general price developments in the crypto market. The dashed dark blue line and the corresponding right-hand scale of Figure 4.2 shows the development of total assets in points of the Crypo Market Index 10 published by SIX (magenta line, right-hand scale). This dashed dark blue line thus shows the development of volumes adjusted for market performance. Analogous to the temporal development of CHF-denominated total assets, a clear trend can also be

[1] Note that some of the products do not exclusively invest directly in crypto assets.

[2] Note that missing values for the funds' monthly total assets in the Morningstar Direct database have, were available, been replaced with data from Bloomberg.

seen here, although the development is steadier. This indicates that new money has continuously flowed into the corresponding investment vehicles over the past months.

As already mentioned, the total size of all ETPs and open-end funds as at the end of April 2022 was around CHF 4.0 billion. Figure 4.3 shows a breakdown of this sum into the following three different perspectives and all of their overlaps:

- **Distribution perspective (in blue):** Includes all ETPs and open-end funds that are available for sale in Switzerland and/or Liechtenstein. This does not preclude these products from also being offered for sale in other countries.
- **Exchange perspective (in green):** Includes all ETPs and open-end funds that are traded on a traditional exchange based in Switzerland.[3] This does not preclude these products from also being traded on exchanges in other countries.
- **Production perspective (in magenta):** Includes all ETPs and open-end funds that are domiciled in Switzerland or Liechtenstein. The term "domicile" refers to the country where the product is legally registered and not necessarily to the country where it is managed.

[3] Note that Liechtenstein does not have a traditional stock exchange.

Figure 4.3 reveals that as of the end of April 2022, the highest number, i.e. 49, and the highest volume of total assets, i.e. CHF 2,258 million, are attributable to ETPs and open-ended funds that are traded, domiciled, and available for sale in Switzerland and/or Liechtenstein. A second large share of the total volume is made up of products that are traded and available for purchase in Switzerland and/or Liechtenstein, but are domiciled in other countries. The ETPs and open-end funds that are relevant in the local crypto assets ecosystem are therefore very often traded in Switzerland, which is why the following sections will focus more on the exchange perspective. The analysis covers activity on the SIX Swiss Exchange and the BX Swiss stock exchanges. The figures for trading in indirect financial vehicles for crypto

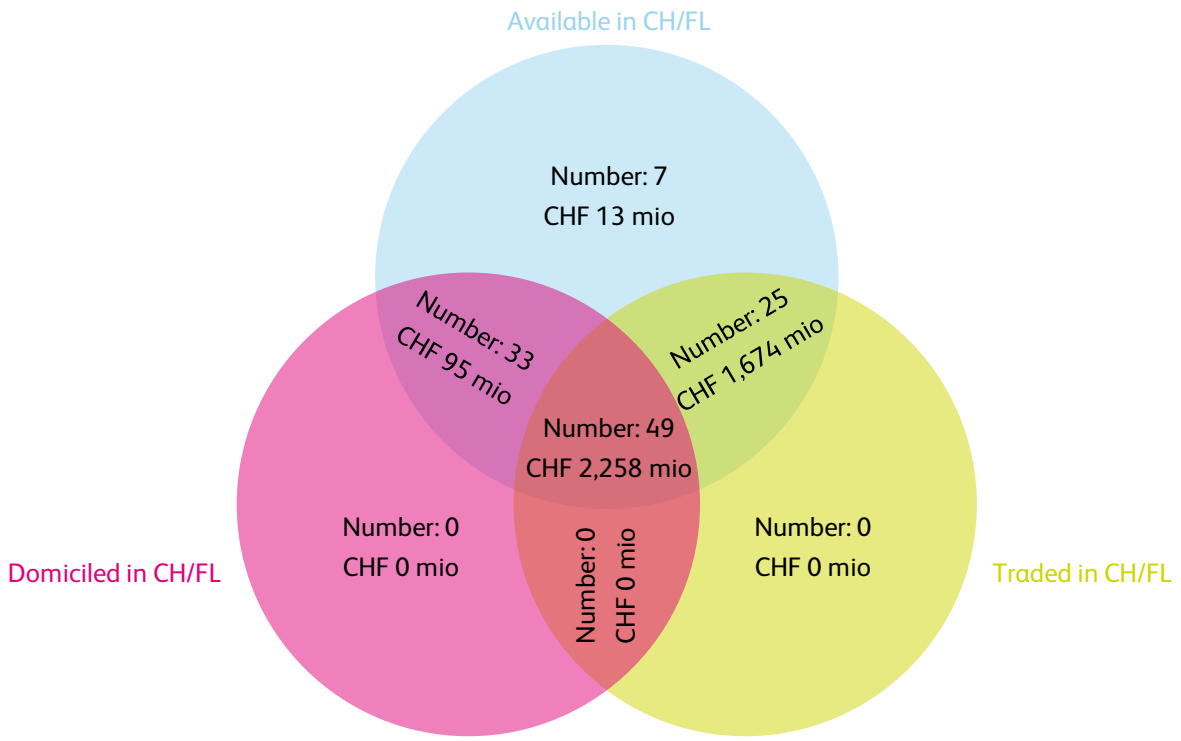

Figure 4.3: ETPs and open-ended funds available, domiciled and/or traded in Switzerland or Liechtenstein (source: Morningstar Direct)

assets on both exchanges were provided directly by the exchange operators or are sourced from their websites.

The temporal development of the number of indirect financial products[4] traded in Switzerland are shown in Figure 4.4, including both ETPs and structured products. The left-hand graph shows the development according to different product types, revealing that ETPs have increased in relevance over time. While 22 products of this type existed in August 2020, the month for which the first data are available, the number of ETPs was already 168 at the end of April 2022. In addition, tracker certificates have also steadily gained in relevance, increasing from 26 to 102 products over the observation period. In contrast, the number of mini futures in particular has decreased from 78 to 17. The decline in mini futures can be explained by the fact that they were closed by the issuer after reaching or exceeding the knock-out threshold. Their number therefore tends to fluctuate to a certain extent with market movements. In general, it is evident that the offer of indirect products for crypto assets has continuously expanded over the last few months. With the introduction of a capital protection certificate with participation on Bitcoin, a new type of structured product also started trading in Switzerland in April 2022. The expansion of the product range is likely to continue in the future. For example, SIX Swiss Exchange has announced that it will launch trading in futures on Bitcoin and Ether in the third quarter of 2022, subject to regulatory approval (SIX, 2022).

The right-hand graph of Figure 4.4 breaks down the indirect financial products according to their underlying. This shows that the number of products on Bitcoin and Ether has remained relatively stable over the observation period, while index products, i.e. investment products on a basket of crypto assets, and products on other crypto assets have gained in relevance. In particular, the strong increase in the latter category is driven by the continuous developments in the field of DLT. For example, recent developments in the area of DeFi and DLT-based metaverses[5] have led to the continuous introduction of products on corresponding crypto assets, e.g. Uniswap or Decentraland. In general, it can be said that the range of underlyings has continuously increased over the past months.

Despite the continuously increasing number of indirect financial products for crypto assets traded in Switzerland in the past months, the corresponding trading volumes have been more volatile, as shown in Figure 4.5. The figure shows the absolute monthly market turnover in million Swiss francs for ETPs and structured products (left-hand graph) and the respective proportional distribution (right-hand graph).

[4] Note that in some cases, multiple financial products are listed under a single ISIN.

[5] See Alioth, Ankenbrand, Bianchi, Bieri, and Kronenberger (2022) for an overview of the current status and developments of metaverses for the financial industry.

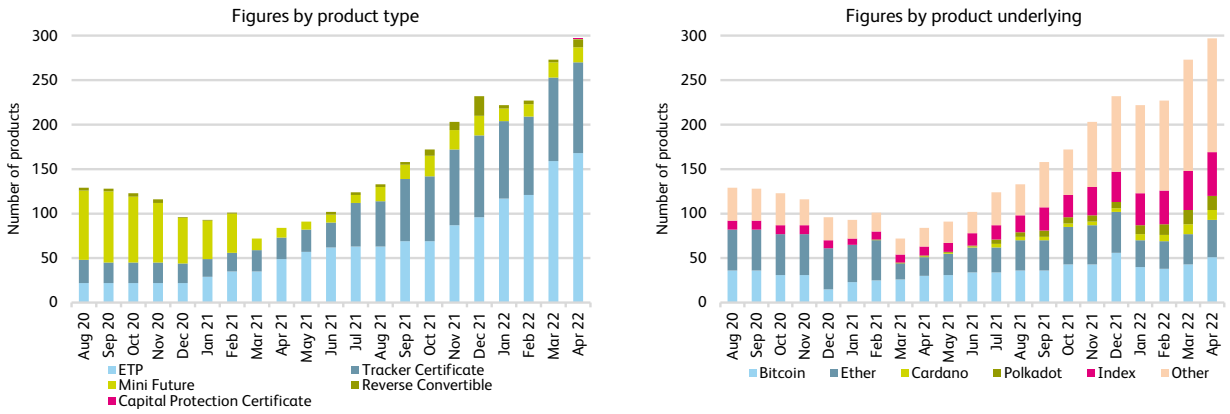

Figure 4.4: Number of crypto-related financial products traded in Switzerland per month by product type (left-hand graph) and underlying asset (right-hand graph) (sources: BX Swiss, SIX)

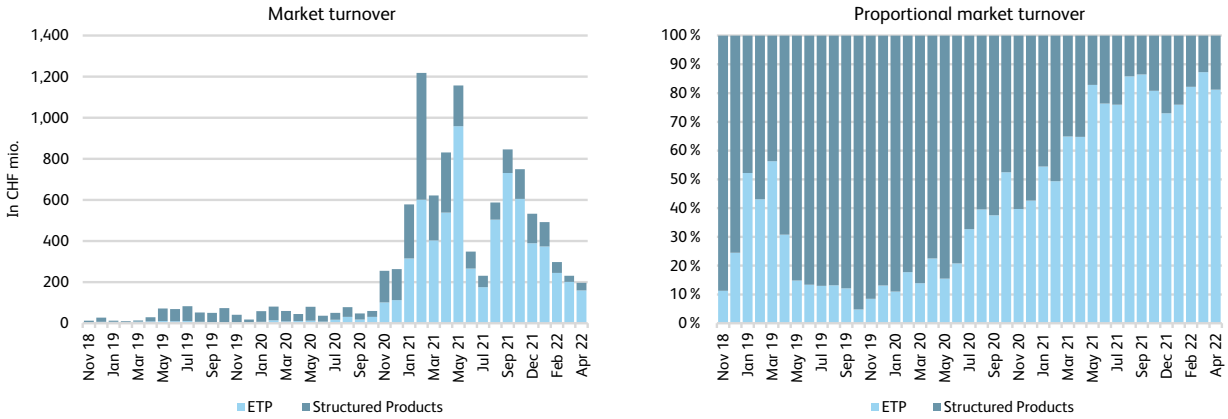

Figure 4.5: Market turnover by month (sources: Bloomberg, BX Swiss, SIX)

The absolute view shows that there were strong fluctuations in the total trading volume and that it decreased continuously towards the end of the observation period. In April 2022, a total volume of CHF 196 million was recorded on the SIX Swiss Exchange and BX Swiss. This is more than six times less than in the peak month of February 2021. With around 80 percent of the total, the majority of market turnover in April 2022 was generated by ETPs, as the right-hand graph of Figure 4.5 shows. The share has risen steadily since the beginning of 2020 and has levelled off at this high level in the most recent months. In the last twelve months of the observation period, a total of CHF 6.5 billion in indirect financial products on crypto assets were traded on the two Swiss exchanges.

Similar findings as for the trading volume can also be found for the number of trades executed for indirect products on crypto assets on Swiss exchanges. The left-hand graph of Figure 4.6 shows that around 13,000 trades were counted in April 2022, which again represents a sharp decline compared to the peak month, May 2021. The largest share of these trades, at over 90 percent, is accounted for by ETPs. As the right-hand graph in Figure 4.6 shows, the proportional distribution of registered trades on ETPs has settled at that high level in recent months, underlining the relevance for the corresponding investment vehicle in the Swiss and Liechtenstein crypto assets ecosystem.

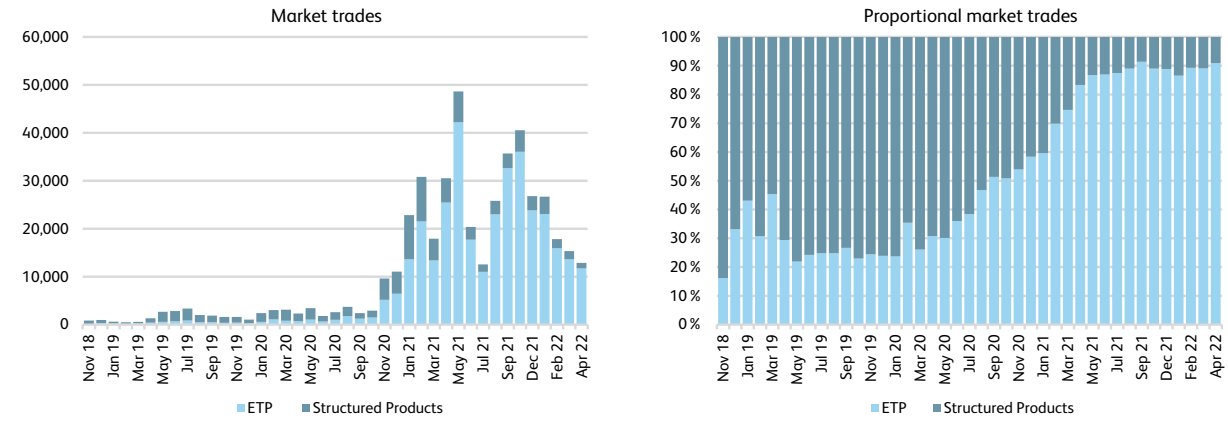

Figure 4.6: Market trades by month (sources: Bloomberg, BX Swiss, SIX)

A breakdown of the base currencies in which the trades are executed can be found in Figure 4.7. The figure shows the clearly dominant role of the US dollar, which accounts for 88 percent of the total market turnover in April 2022. The second most popular currency is the Swiss franc (7 %), followed by the euro (3 %) and the British pound (2 %). The Japanese yen, for which a trading volume was recorded for the first time in March 2022, plays a negligible role. Figure 4.7 also shows that the proportions between the currencies have changed only marginally since the launch of the first indirect financial product on crypto assets in Switzerland in November 2018.

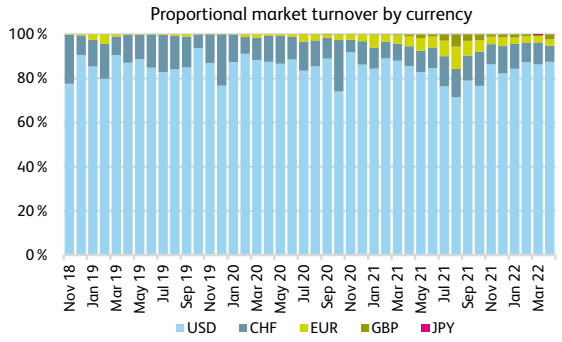

Figure 4.7: Proportion of turnover by currency by month (sources: Bloomberg, BX Swiss, SIX)

4.2. Direct Investments

In contrast to indirect investments, crypto assets can also be traded directly on centralised and decentralised crypto exchanges. In the following, therefore, the corresponding trading volume originating from Switzerland is discussed. The data for this is collected using the API provided by CoinGecko (2022a) and focuses on the period from May 1, 2021 to April 30, 2022. Firstly, the daily spot trading volumes of all centralised and decentralised exchanges available via the API (n=434) were retrieved for the specific period. Secondly, the daily trading volumes denominated in Bitcoin were multiplied with the corresponding day's BTC/CHF market rate and aggregated by the period. Thirdly, the 15 largest exchanges per segment were selected for further analysis. In regard of centralised exchanges, however, only exchanges with a CoinGecko trust score greater than five out of ten were considered. The trust score focuses on liquidity, scale of operations, and API coverage CoinGecko (2022b). Finally, based on the assumption that the Swiss web traffic of an exchange reflects the effective Swiss trading volume on the exchange, the web traffic data of each exchange was collected using the platform of the search engine marketing company Semrush (2022) and multiplied by the total trading volume of the exchange to determine the volume originating from Switzerland. As some exchanges are frequently accessed via mobile application, the Swiss market shares have to be considered with care. Additionally, the estimations of the Swiss market shares might be rather modest given that Switzerland is a wealthy country and that Swiss investors might hold an over-proportional amount of crypto assets compared to their web traffic.

Centralised Crypto Exchanges

As listed in Table 4.1, the total trading volume for direct investments in crypto assets on centralised exchanges is estimated at CHF 81.2 billion for Switzerland in the 12-month observation period. The crypto exchange Binance accounts for roughly 34 percent of the total trading volume of the 15 largest centralised crypto exchanges and also for 46 percent of the estimated Swiss trading volume. The Swiss web traffic share of Binance is therefore higher than the average Swiss web traffic share of all platforms. In terms of web traffic, the three exchanges FTX, CoinFLEX, and Kucoin have the highest traffic rates and are therefore accessed the most from Switzerland in relative terms. Most exchanges based in or operated from Asian countries such as Upbit, XT.COM, BKEX, and ZB report comparably lower Swiss web traffic shares.

Decentralised Crypto Exchanges

Analysing the trading volumes on decentralised crypto exchanges in Table 4.2, different findings emerge. Both the total volume originating from Switzerland and the average share of Swiss web traffic are lower compared to centralised exchanges, amounting to CHF 5.1 billion and 0.24 percent, respectively. The majority of the Swiss trading volume is generated by the exchanges Mdex, Pancakeswap (v2) and Solarbeam. The platforms Solarbeam and Quickswap report the highest Swiss web traffic shares.

Comparing the Swiss trading volume of all 15 decentralised exchanges to the total trading volume at centralised exchanges, it seems that Swiss traders appreciate the advantages offered by centralised exchanges such as user friendliness, reliability in transaction processing, and higher liquidity.

Derivatives Crypto Exchanges

In addition to spot trading, some centralised crypto exchanges have increasingly started to provide indirect investment products such as derivatives on crypto assets. Table 4.3 shows that the total corresponding derivative trading volume originating from Switzerland is estimated at CHF 119 billion. Binance (Futures) is processing 45 percent of the volume of the 15 largest derivatives crypto exchanges. Nevertheless, according to the estimated Swiss trading volume, the exchanges FTX (Derivatives) and Bybit seem to be the preferred platforms for Swiss investors, with estimated annual trading volumes of roughly CHF 36 billion and CHF 32 billion, respectively. In contrast, the derivatives exchanges KuCoin Futures, Hopex, and BTCEX (Futures) are not used by Swiss investors.

It is noteworthy that the total derivatives trading volume exceeds the total spot trading volume at centralised and decentralised exchanges. However, as for derivatives markets of traditional assets, there is probably a large difference in volume as a proxy of the notional value, which indicates the total value of underlying assets, and the actual market value of the derivatives set by supply and demand.

Regarding the product offering of the 15 largest derivative exchanges, 1,830 perpetual pairs and 635 futures pairs are reported (CoinGecko, 2022a). The majority of futures pairs are offered by the two platforms OKX (Futures) and FTX (Derivatives), accounting for 84 percent of all futures pairs. Perpetual contracts, which are cash settled and, in contrast to traditional futures, do not have an expiration date, are issued by and traded on each of the 15 largest derivative exchanges for crypto assets.

Although the 15 largest derivative exchanges listed in Table 4.3 are all operated by centralised providers, there are examples of decentralised exchanges, such as dYdX Perpetual, which offer at least partly decentralised derivative trading. On the path to a fully decentralised platform, dYdX Perpetual is for example focusing on challenges like decentralising the currently centrally run orderbook and its matching engine, which in the past enabled the exchange to scale their trading volumes efficiently (dYdX, 2022).

Table 4.1: Annual trading volumes of the 15 largest centralised crypto exchanges, May 1, 2021 to April 30, 2022 (sources: CoinGecko, Semrush)

Platform Name	Total Trading Volume in CHF mil.	Traffic Share CH in %	Trading Volume CH in CHF mil.
Binance	8,151,290	0.46	37,108
Upbit	2,240,646	0.001	27
OKX	2,127,750	0.16	3,439
Huobi Global	1,721,131	0.22	3,851
Coinbase Exchange	1,449,484	0.55	7,911
FMFW.io	1,109,439	0	0
Bitrue	1,105,949	0.43	4,757
CoinFLEX	880,104	0.93	8,175
XT.COM	815,828	0	0
Crypto.com Exchange	763,891	0.39	2,984
Digifinex	744,193	0.09	685
KuCoin	737,888	0.71	5,225
FTX	735,871	0.96	7,040
BKEX	652,694	0	0
ZB	549,761	0.004	21
Total	**23,785,920**	**0.34**	**81,223**

4.3. Tokenisation

Tokenisation describes the process of issuing tokens on a blockchain. One form of tokenisation is the issuance of stablecoins, i.e. DLT-based tokens that are pegged to the value of another asset such as a traditional fiat currency. A selection of three stablecoins that are pegged to the Swiss franc is listed in Table 4.4. The one with the largest number of transfers (38,504) and holders (825) at the end of the second quarter of 2022 is the stablecoin JCHF by Jarvis Network which is available on multiple networks. Bitcoin Suisse's Ethereum-based XCHF is the stablecoin with the largest circulating supply of 3.1 million tokens, equivalent to CHF 3.1 million. Another Ethereum-based stablecoin is sCHF from Synthetix, accounting for 1,584 transfers, 92 holders, and a circulating supply of roughly 1.4 million tokens.

Tokenisation goes beyond stablecoins, however, with a multitude of providers (see Figure 3.2) focusing on different assets. A survey by Brunner (2022) shows that the number of token generating events by Swiss providers has increased in 2021 compared to previous years. The 15 institutions that issued tokens and participated in the survey indicated that they issued a total of eight non-security tokens, i.e. payment or utility tokens, and a total of 60 security tokens (e.g. tokenised shares) for 2021 (Brunner, 2022).

There was also activity in the area of issuing non-fungible tokens (NFTs), i.e. unique cryptographic tokens that are not interchangeable compared to fungible tokens[9] such as units of Bitcoin, in Switzerland in 2021. For example, the Swiss Post partnered with Inacta to issue 175,000 digital crypto stamps for CHF 8.90 per token on the Polygon network (Swiss Post, online). Another example is the tokenisation of the painting "Fillette au béret" by Picasso by the Swiss crypto bank Sygnum in partnership with the art investment company Artemundi. This fractional ownership is intended to lower financial barriers and democratise access to works of art history (Sygnum, 2021).

4.4. Asset Management

Asset management for crypto assets is a service provided for direct investments, but also for indirect investment products, as part of their risk can be hedged by

[9] For a classification framework for different types of (crypto) assets, see Ankenbrand, Bieri, Cortivo, Hoehener, and Hardjono (2020).

Table 4.2: Annual trading volumes of the 15 largest decentralised crypto exchanges, May 1, 2021 to April 30, 2022 (sources: CoinGecko, Semrush)

Platform Name	Total Trading Volume in CHF mil.	Traffic Share CH in %	Trading Volume CH in CHF mil.
Uniswap (v3)	583,822	0.06	361
Mdex	325,242	0.42	1,376
PancakeSwap (v2)	212,459	0.69	1,456
Uniswap (v2)	191,489	0.06[6]	119
Sushiswap	121,709	0.01	15
Solarbeam	97,910	1.01	994
Aave	81,754	0.10	83
Mdex BSC	78,241	0	0
Trader Joe	77,519	0.41	318
Sushiswap (Arbitrum One)	76,891	0.01[7]	10
Curve Finance	76,153	0.08	63
Pangolin	57,704	0	0
Quickswap	47,947	0.73	349
Raydium	41,098	0	0
Serum DEX	39,719	0	0
Total	**2,109,657**	**0.24**	**5,143**

[6] Traffic according to root-domain (analog Uniswap (v3)).
[7] Traffic according to root-domain (analog Sushiswap).

Table 4.3: Annual trading volumes of the 15 largest derivatives crypto exchanges, May 1, 2021 to April 30, 2022 (sources: CoinGecko, Semrush)

Platform Name	Total Trading Volume in CHF mil.	Traffic Share CH in %	Trading Volume CH in CHF mil.
Binance (Futures)	22,646,461	0.02[8]	3,554
OKX (Futures)	7,297,533	0.16	11,794
FTX (Derivatives)	3,778,553	0.96	36,147
Huobi Futures	3,774,078	0.22	8,445
Bybit	3,493,719	0.92	32,042
Bitget Futures	1,929,577	0.01	119
CoinFLEX (Futures)	1,611,657	0.93	14,971
KuCoin Futures	826,347	0[8]	0
Gate.io (Futures)	816,716	0.18	1,451
Hopex	803,124	0	0
Phemex (Futures)	771,834	0.45	3,466
Bitforex (Futures)	720,061	0.03	225
BTCEX (Futures)	713,561	0	0
BitMEX (Derivative)	615,396	0.82	5,071
Prime XBT	560,651	0.32	1,811
Total	**50,359,267**	**0.24**	**119,096**

[8] Traffic according to futures specific sub-domain.

investing directly in crypto assets on which the issued product is based. The latter is reflected, for example, in the balance sheet of Bank Vontobel, which held crypto assets at a fair value of CHF 241.0 million at the end of 2019. The strong growth of the indirect crypto asset market was also reflected in the growth of the fair value of crypto assets in the Bank's balance sheet, which was already CHF 985.5 million at the end of 2020, representing an increase of CHF 744.6 million or a relative growth rate of 309 percent year-on-year. However, at the end of 2021, the fair value of crypto assets decreased by CHF 416.5 million to CHF 569 million, a decline of more than 42 percent year-on-year (Vontobel, 2021).

21Shares, another provider of indirect crypto asset products, held a total of USD 39.8 million in crypto assets at the end of 2019, which grew to USD 284.9 million by the end of 2020. As of the end of 2021, 21Shares recorded USD 2,324.8 million in crypto assets, an increase of USD 2,039.9 million in absolute terms or a year-on-year growth rate of 816 percent (21Shares, 2021).

Leonteq, an issuer of structured products, does not explicitly show the amount of crypto assets in its balance sheet. However, the company states that outstanding volumes in products with crypto assets as underlying assets have increased from CHF 23 million at the end of 2019 to CHF 155 million at the end of 2020 to CHF 487 million at the end of 2021, an increase of CHF 132 million (from 2019 to 2020) and CHF 332 million (from 2020 to 2021), or 574 percent and 214 percent in relative terms, respectively (Leonteq, 2021).

Table 4.4: Selection of CHF-stablecoins, end of June 2022 (sources: Blockscout, BscScan, Etherscan, PolygonScan, SnowTrace)

Stablecoin	Transfers	Holders	Token supply	Network(s)	Issuer
JCHF	38,504	825	1,963,584	Avalanche, BNB Chain, Ethereum, Gnosis, Polygon	Jarvis Network
XCHF	11,724	374	3,100,000	Ethereum	Bitcoin Suisse
sCHF	1,584	92	1,361,668	Ethereum	Synthetix

4.5. Custody

Custody of crypto assets aims to keep crypto investments secure and accessible. However, unlike physical wallets, crypto asset custodians primarily guard a user's private keys rather than the assets themselves, as the associated data and transactions are stored on a blockchain. An investor may choose to self-custody in order to personally manage the private keys, although in this case he or she must also bear all the associated risks. Since losing access to a crypto wallet or possible hacks means losing the assets, third-party custody has become a viable option for many investors. Providers of custody services are exchanges, digital asset managers, and custodial banks (Coindesk, 2022).

For the latter two types of custodians, this year's survey identified eleven participating companies. When comparing the customer segments of the companies offering custody services as listed in Figure 3.1, family offices and other institutional clients are targeted by 91 percent of the companies. However, only 27 percent of the companies offer custody solutions for retail clients.

Many custodians also offer smart custody solutions for example by staking assets to generate an additional income for their clients. For some blockchain networks such as the Ethereum Beacon Chain, which introduces the Proof-of-Stake consensus mechanism for Ethereum, information about staking deposits can be obtained in dashboards provided by community members of the platform Dune Analytics.

For the Ethereum Beacon Chain specifically, Bitcoin Suisse ranks as the fourth largest known deposit provider, reaching a share of approximately 2.21 percent with ETH 288,096, equivalent to CHF 329 million, as of July 4, 2022 (Dune Analytics, 2022).

In general, centralised crypto exchanges also offer custody solutions so that their customers do not have to maintain their own wallet. When investigating volumes of assets under custody at crypto exchanges, information about exchange reserves can be obtained by crypto asset from the data provider Crypto Quant (2022). Taking into account exchange reserves for Bitcoin and Ether, the largest 15 centralised exchanges, as reported in Table 4.1, hold around 55 percent of the reserves of all crypto exchanges.[10] The exchange with the largest reserve is Coinbase Pro with 29 percent, followed by Binance with 18 percent and OKX with 2.9 percent, which is equivalent to CHF 45.6 billion, CHF 28 billion, and CHF 2.7 billion, respectively. Considering the exchange reserves as a proxy for assets under custody, the Swiss assets under custody can be estimated using the web traffic share in Table 4.1. This results in estimated total assets of CHF 426 million which are held by Swiss investors on centralised crypto exchanges. As for all exchange reserve volumes, Coinbase Pro (58 %) and Binance (30 %) hold the highest estimated shares of Swiss assets under custody. Here it also becomes apparent that the previously estimated trading volumes do not necessarily lead to a proportional share in assets under custody.

The estimation of assets under custody for decentralised exchanges is not available, as users take care of custody themselves.

[10] Only for eight of the 15 largest centralised exchanges, exchange reserves are reported in BTC and ETH. Three exchanges lack the exchange reserve volume for one of the two crypto asset in consideration.

5. From Crypto Assets to DeFi

In this chapter, an overview of crypto assets from a Swiss investor's perspective is provided (Section 5.1) and developments in the field of decentralised finance (DeFi) are discussed (Section 5.2).

5.1. Crypto Assets as an Investment

Since the inception of the Bitcoin network originally described by Nakamoto (2008) in the year 2009, the space for crypto assets has developed rapidly with the market being subject to different cycles. After a sharp increase in total market capitalisation towards the end of 2017, the market cooled down in 2018 and 2019, as shown in Figure 5.1. At the end of 2020 and the beginning of 2021, total market capitalisation has risen sharply again. In mid-2021, there was a comparatively short correction, followed by a rise to the all-time high of roughly USD 2.7 trillion in November 2021. Towards the end of the observation period, the total capitalisation of the market for crypto assets decreased again and stood at around USD 1.5 billion at the end of April 2022 (Coin Metrics, 2022).

Figure 5.1: Total capitalisation of the market for crypto assets (source: Coin Metrics (2022))

Crypto assets can themselves be considered an asset class and make a useful contribution as a portfolio addition to investment management (Ankenbrand & Bieri, 2018). The basis for this is the (historically) low return correlation of crypto assets with other asset classes. An analysis of the rolling 30-day correlation between Bitcoin and the Swiss Performance Index® (SPI) is shown in Figure 5.2.

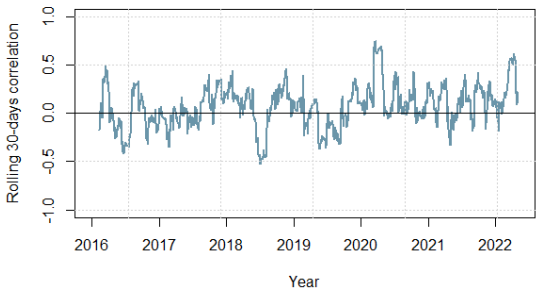

Figure 5.2: Rolling 30-days return correlation between Bitcoin and the SPI (source: Bloomberg)

It can be seen that the rolling return correlation since the beginning of 2016 has mostly moved in the range between -0.5 and 0.5. Bitcoin can therefore be seen as uncorrelated to the stock market, implying a certain potential for portfolio diversification. The two largest values are reached in March 2020, at the outbreak of the Covid-19 crisis, i.e. during a phase in which both Bitcoin and the SPI lost considerable in value, and in March 2022.

The potential of crypto assets, as proxied by Bitcoin, for portfolio diversification for a Swiss investor is discussed in the following. Considering the asset classes stocks (as proxied by the Swiss Performance Index®), bonds (as proxied by the Swiss Bond Index® TR), real estate (as proxied by the SXI CH Real Estate® Shares TR), and Bitcoin (denominated in CHF), two different portfolio allocations as defined in Table 5.1 are compared regarding their performance. The first portfolio is based on the overall investment strategy of Swiss pension funds according to the Occupational Pension Supervisory Commission (OPSC) and serves as a benchmark (OPSC, 2020). Using this strategy, 40 percent is invested in bonds, 35 percent in equities, and 25 percent in real estate.[1] The second portfolio additionally includes Bitcoin. More specifically, exposure to traditional asset classes is reduced by 1 percentage point

[1] In this analysis, alternative investments are not included in the asset universe.

Table 5.1: Asset allocations considered

Portfolios	Bonds	Stocks	Real Estate	Bitcoin
Portfolio excluding BTC	40 %	35 %	25 %	0 %
Portfolio including BTC	39 %	34 %	24 %	3 %

each and newly invested in Bitcoin, resulting in a 3 percent investment into the latter.[2]

Despite this relatively low portfolio weight of Bitcoin, the addition of a crypto asset, considering yearly rebalancing, results in a outperformance of the portfolio compared to the benchmark portfolio over the examination period (January 2016 to April 2022). The cumulative returns of the two portfolios are illustrated in Figure 5.3, along with the temporal development of corresponding maximum drawdown values, a measure of a portfolio's largest value drop from a peak to a trough.

Although the inclusion of a high-risk asset such as Bitcoin led to an outperformance, the maximum drawdowns of the two portfolios do not differ substantially. Over the whole observation period, the largest (temporary) loss for both portfolios was about 18 percent and did occur in the first quarter of 2020, which was characterised by a Covid-19-related market correction. However, between the beginning of 2018 and the end of 2019, the maximum drawdown values for the portfolio including Bitcoin are considerably higher than those of the benchmark portfolio due to the Bitcoin crash at the beginning of 2018. The annualised figures for the returns, the associated risk (measured by the standard deviation), and the Sharpe ratios[3] for both portfolios are listed in Table 5.2.

It shows that the portfolio including Bitcoin was subject to higher fluctuations than the benchmark portfolio, as measured by the larger standard deviation, but also achieved a higher annualised return of 12.1 percent compared to 5.4 percent. The Sharpe ratio, as a measure of risk-adjusted performance, illustrates that an investment in Bitcoin, albeit with 3 percent of total assets a comparably small one, has already led to

[2] This is a more conservative allocation compared to the market portfolio according to Modern Portfolio Theory, which retrospectively results in an optimal Bitcoin share of over 20 percent over the same observation period.

[3] Note that a risk-free rate of 0 percent is assumed.

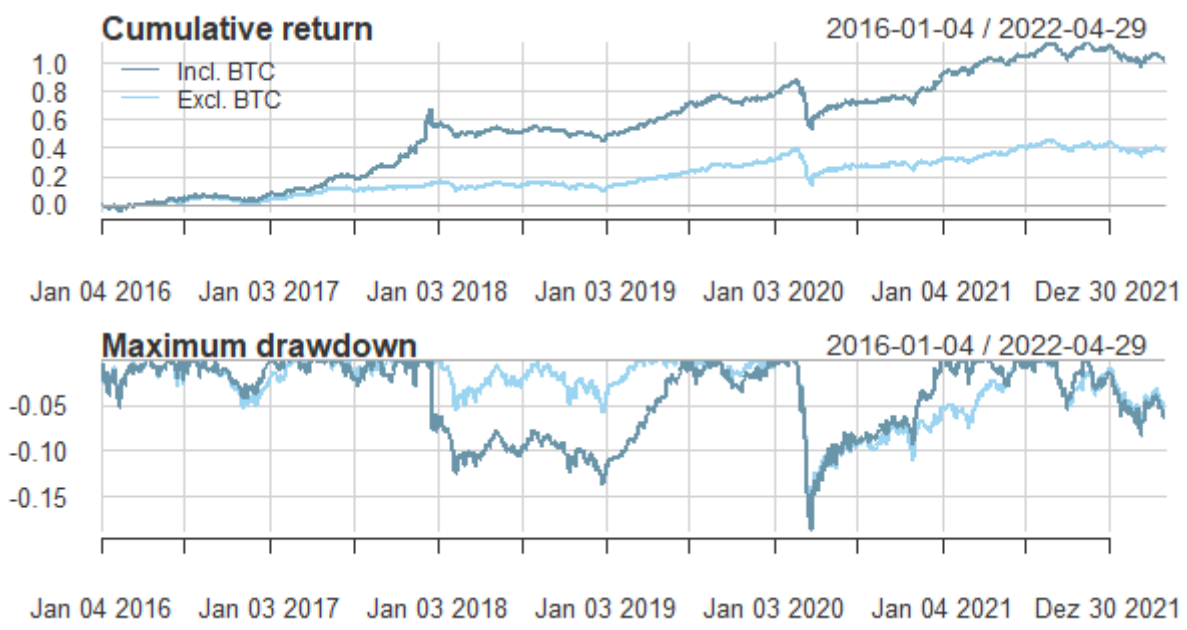

Figure 5.3: Portfolio performances in- and excluding Bitcoin

Table 5.2: Portfolio performances in- and excluding Bitcoin

Portfolios	Return	Std dev	Sharpe ratio
Portfolio excluding BTC	5.4 %	7.4 %	0.73
Portfolio including BTC	12.1 %	9.1 %	1.33

an improvement in risk-adjusted performance from the perspective of a Swiss investor.

Note that in the simplified case described, only Bitcoin is considered as a possible investment from the new asset class of crypto assets, and that other factors such as reputation, security, and regulation are important for investment management in addition to the quantitative measures provided. It should also be noted that past performance is no guarantee of future performance.

In general, it can be stated that the continuing technological innovation in the underlying distributed ledger technology has consequently led to an increasing transformation of traditional financial markets into crypto-based markets and to the emergence of a broad range of crypto assets, some of which go beyond the functionality of Bitcoin. In this context, an increased observation of asset tokenisation can be observed. Through tokenisation (see Section 4.2), a DLT-based digital wrapper for bankable assets such as stocks or bonds, tangible assets such as arts or real estate, or intangible assets such as patents and copyrights is created. Tokenisation of assets can thus serve as a wrapper for new investment opportunities, similar to traditional financial products such as structured products and funds where performance is tied to an underlying asset. However, these new crypto assets can differ significantly from traditional assets in certain attributes such as the underlying technology, the information complexity, or the way consensus on the state of the asset is reached. This poses a challenge to the various stakeholders such as investors or regulators in retaining an overview of existing assets of different types and, in particular, of their design and individual characteristics. A holistic framework to address this challenge is presented in Ankenbrand et al. (2020). Their proposed taxonomy can be used to evaluate any type of asset in a structured way and takes into account the increasing fusion of traditional finance with the crypto assets ecosystem.

5.2. Decentralised Finance

The merging of traditional finance with decentralised technologies is often summarised under the term "Decentralised Finance", or "DeFi" for short. This development creates new use cases in banking beyond tokenisation. DeFi solutions are typically designed to mimic traditional financial services but are based on trustworthy and transparent systems that do not require intermediaries (e.g. banks or exchanges), and are thus in principle accessible without restrictions.[4] The openness of the system is therefore at least partially comparable with the goals of Open Finance from the traditional financial world. One of the main differences to the "Open Finance" approach is that the booking of assets is no longer done by financial institutions but by the underlying distributed ledger technology.[5] The products and services offered are based on smart contracts, where predefined rules are enforced automatically and autonomously, and all associated data is stored in a decentralised manner through a distributed ledger (e.g. a blockchain). Currently, the most widespread DeFi solutions can be found in the areas of trading, peer-to-peer payments, insurance, the issuance of stablecoins, and lending (Bitcoin Suisse, 2020). The size of the market for such decentralised banking applications has increased rapidly over the past months, as shown in Figure 5.4. When the first DeFi applications were introduced at the end of 2018, the total volume locked in corresponding smart contracts was lower than USD 1 billion until around mid 2020. Thereafter, the volume increased rapidly, peaking in December 2021 at around USD 315 billion (DeFiLlama, 2022). This rapid development of the DeFi ecosystem and its continuous expansion of the range of services offered is also relevant for the asset financial industry, as these solutions provide a new channel for certain products and services without the need of intermediaries.

[4] For a more detailed overview on DeFi, see Schär (2021).

[5] A more detailed comparison of Open Banking, Open Finance, and DeFi can be found in Ankenbrand, Bieri, Frigg, Grau, and Lötscher (2021).

Features such as full transparency, the elimination of trustees or custodians, and the automation enabled by the underlying DLT infrastructure have already led to a range of products that offer various ways to track, manage, or hedge assets, and these solutions increasingly include traditional investments in tokenised form (e.g. tokenised shares).

DeFi developments should therefore be clearly monitored by the financial industry in order to identify potential opportunities as well as risks at an early stage. However, the direction in which the ecosystem will develop and the extent to which, or the speed at which, the traditional financial world will merge with DLT also depends to a large extent on technological progress in this area and on the regulatory environment. In addition, recent developments in the DeFi space, such as the (temporary) suspension of certain platforms due to liquidity issues, have shown that investors should be very aware of the risks associated with these novel products and services.

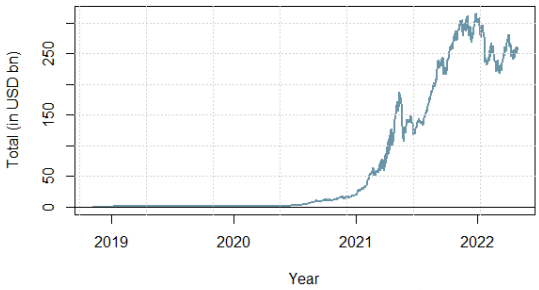

Figure 5.4: Total value locked in DeFi (source: DeFiLlama (2022))

6. Conclusion and Outlook

The second edition of the "Crypto Assets Study" provides an overview of the current status and developments in the Swiss and Liechtenstein crypto assets investment ecosystem and continues the facts and figures on business activity collected for the first time last year. The findings are summarised in the following statements and theses:

The Swiss and Liechtenstein ecosystem for crypto assets investments is vibrant. Despite (or maybe because of) high volatility, temporarily rising return correlation with other asset classes, hypes, and crashes, the Swiss and Liechtenstein ecosystem for crypto assets investments is still active. In general, the development can be described as relatively stable compared to the results of last year's edition of this study.

New products enter the market. The Swiss investment ecosystem for crypto assets investments is offering more and more financial products. In the area of indirect investments, this can be seen, among other things, in the number of ETPs that are domiciled, traded, or available for sale in Switzerland and/or Liechtenstein. Not only is the number of products increasing, but also their diversity. For example, the range of investable underlyings of indirect financial products is expanding, and new product types are also being introduced. The latter is exemplified by the planned introduction of futures contracts on Bitcoin and Ether on the SIX Swiss Exchange. With the development of the ecosystem, further innovations can be expected, not only in the area of indirect investments but also for direct investments, for example driven by the maturation of decentralised finance.

Broad product and service offering, but not for all customer segments. The ecosystem offers both direct and indirect investment solutions in crypto assets, as well as infrastructure services. In this context, the majority of companies surveyed target corporate and/or institutional rather than retail customers. Issuance and tokenisation solutions as well as exchange services are the most frequently mentioned offerings. Lending services, in contrast, are provided by comparatively few companies. The evaluation furthermore shows that there are specialists in the ecosystem who only offer individual services of the entire value chain of crypto assets investments, but also few generalists with a broader offering across the entire investment value chain.

Volumes on centralised crypto exchanges remain the largest. Crypto assets or corresponding derivatives can be traded on traditional exchanges as well as on centralised or decentralised crypto exchanges. The largest trading volume is transacted on centralised crypto exchanges. More specifically, the derivatives and spot trading volume originating from Switzerland on centralised crypto exchanges amounts to an estimated CHF 119 billion and CHF 81 billion, respectively. The trading volume on decentralised crypto exchanges amounts to CHF 5.1 billion in the last twelve months of the observation period and is thus smaller than that on traditional exchanges, which account for CHF 6.5 billion.

High risks, but also potentially high profits. An introductory analysis shows that even simple investment strategies in crypto assets could have been beneficial for investors in the past by delivering higher risk-adjusted performance in a portfolio context. This is, however, associated with increased risk, as evidenced by standard risk measures such as the standard deviation or the maximum drawdown of a portfolio including crypto assets. Moreover, in times of high volatility, the return correlation with traditional investments can be high, which temporarily reduces the diversification potential of crypto assets.

7. Factsheets of Companies in the Swiss and Liechtenstein Crypto Assets Ecosystem

The last chapter of this study contains the factsheets of the companies that participated in the survey. At this point, we would like to thank all companies that supported our initiative to portray the Swiss and Liechtenstein ecosystem for crypto assets investments in a most comprehensive way.

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21Shares AG

https://21shares.com/

21SHARES

21Shares is the world's largest issuer of cryptocurrency exchange-traded products (ETPs) and is committed to making cryptocurrency investing more accessible. In 2018, the company listed the world's first physically backed crypto ETP.

Domicile	CH	**Year of Inception**	2018	**Valuation**	
Employees … of which in CH	**130** 40	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Hany Rashwan, Ophelia Snyder, Lucy Reynolds				
Board Members	Hany Rashwan, Ophelia Snyder, Cathie Wood				
Key Partners	Exchanges (SIX Swiss Exchange, Deutsche Borse Xetra, Euronext Amsterdam/Paris, etc), Bank/Brokers (Swissquote, TradeDirect, comdirect, InteractiveBrokers, etc), Custodians (Coinbase, Copper, Bitcoin Suisse, etc.)				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

4cash.exchange - 4bridges GmbH

https://4cash.exchange - https://4bridges.ch

4bridges™

Swiss based Cryptocurrency Exchange & Community. We follow the cryptocurrency principals & share the values of the cryptocurrency community. Great team of open minded, highly motivated entrepreneurs. With the launch of 4cash.exchange important infrastructure has been created, licensed, and registered for the community. Access to financial freedom and self-custody has become reality.

Domicile	ZH	**Year of Inception**	2019	**Valuation**	CHF 8,000,000
Employees … of which in CH	**5** 2	**Total Funding** … in 2021	**CHF 605,000** CHF 60,000	**Date of last Funding** State of last Funding	**01.11.2021** Seed
Management Team	Simon Fundel, Emad Hassanipanah, Robin Caduff				
Board Members	Simon Fundel, Emad Hassanipanah, Robin Caduff				
Key Partners	MME Legal AG, KYC Spider AG, Postfinance				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Aktionariat AG

https://aktionariat.com/

Aktionariat

Aktionariat AG offers a set of tools for Swiss companies to create a market for their shares on their own website. Open technology. No intermediaries. Powered by the Ethereum blockchain.

Domicile	ZH	**Year of Inception**	2020	**Valuation**	CHF 13,800,000
Employees … of which in CH	**13** 11	**Total Funding** … in 2021	**CHF 1,700,000** CHF 1,200,000	**Date of last Funding** State of last Funding	**ongoing, cont. fundraising** Seed
Management Team	Murat Ögat, Luzius David Meisser				
Board Members	Murat Ögat, Luzius David Meisser				
Key Partners	LEXR, Monerium, Bity, Konsento				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

Arab Bank (Switzerland) Ltd

https://www.arabbank.ch/

ARAB BANK SWITZERLAND

Arab Bank Switzerland is a bridge between the Arab and Western world. For over 50 years we have championed business leaders and family entrepreneurs.

Domicile	GE	**Year of Inception**	1962	**Valuation**	
Employees … of which in CH	**120** 120	**Total Funding** … in 2021	**CHF 575,000,000**	**Date of last Funding** State of last Funding	
Management Team	Serge Robin, Rani Jabban, Jean-Pierre Daccache, Mark De Snaijer, Richard Rochat, Imed Khammari, Rachid Jami, Stéphane Sépède, Patrick Pillon, Jean-Yves Mathey-Doret				
Board Members	Pierre-Olivier Fragnière, Claude Suchet, Jean Pierre Roth, Wahbé Tamari, Gérard Lohier, Omar Kamal, Alessandro Cosimo Héribert Bizzozero				
Key Partners	Taurus, Kraken, B2c2, OSL, Woorton, Bitstamp, Circle.				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Atomyze AG
https://www.atomyze.ch

ATOMYZE

Our vision is to bring new access, increased liquidity, optimized efficiency, and improved transparency to the commodities industry through Tokenisation and Distributed Ledger Technology (DLT) within a seamless ecosystem. Tokenisation technology can bring flexibility and optionality to the commodities market, bringing investment and industrial worlds together. We also envision ability to tokenize other physical assets in the future.

Domicile	ZG	**Year of Inception**	2018	**Valuation**	n/a
Employees … of which in CH	**over 40** over 40	**Total Funding** … in 2021	**n/a** n/a	**Date of last Funding** State of last Funding	**n/a** n/a
Management Team	Marco Grossi (CEO), Philipp E. Dettwiler (COO), Valerio Matriciani (CRO), Sibil Melliger (CLO), Michael Stockinger (CFO), Bertalan Vecsei (CTO)				
Board Members	Alexander Stoyanov, Konstantin Graf von Schweinitz, Alexander Freedland, Robert Osborne, Stephan Arnet				
Key Partners	Mining companies, banks, warehouses, industrial companies, manufacturing companies, custodians, market makers, exchanges, other raw materal producers, investment houses, trading companies				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

aXedras AG
https://www.axedras.com/

aXedras

aXedras is a fast growing and dynamic DLT / blockchain-based infrastructure and application provider (headquartered in Switzerland – an international hub for both DLT / blockchain solution providers and the precious metals industry) with the vision to connect and digitalize the precious metal industry. aXedras has developed a distributed ledger application for product and data integrity within the bullion market: the Bullion Integrity Ledger™. aXedras combines bullion market expertise and cutting-edge information technology to create significant benefits for the stakeholders of the precious metals industry.

Domicile	SZ	**Year of Inception**	2019	**Valuation**	
Employees … of which in CH	**27** 14	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	**01.05.2022** Series B
Management Team	Urs Röösli, Iwan Lottenbach				
Board Members	Urs Röösli, Frank Suess, Oliver Kehl, Alex Seidel, Florian Herzog, David Tait				
Key Partners	World Gold Council (WGC), Swiss Association of Manufacturers and Traders in Precious Metals (ASFCMP), Swiss Better Gold (SBG), Microsoft, r3 Ltd., ScaleFocus, Swiss Finance and Technology Association				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Base58 Capital AG
https://base58.ch/

We are a technology-driven investment firm specialised in crypto assets.

Domicile	ZG	**Year of Inception**	2017	**Valuation**	
Employees … of which in CH	**3** 2	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Christian Dominik Frey, Ivo Sauter, Sothy C. Kol, Tomasso Bonanata, Fabio Federici				
Board Members	Christian Dominik Frey, Ivo Sauter, Fabio Federici				
Key Partners	Coinbase Prime, Fireblocks, FTX				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Berner Kantonalbank AG
https://www.bekb.ch/

The Berner Kantonalbank BEKB (Bernese cantonal bank) is strongly rooted in the Bern and Solothurn regions. It gives out loans and manages assets, making an important contribution to the independent development of the city of Bern and playing a substantial economic and social role.

Domicile	Switzerland (Bern)	**Year of Inception**	1834	**Valuation**	CHF 100,000,000
Employees … of which in CH	**1,014** 1,014	**Total Funding** … in 2021	**CHF 1,127,000,000** CHF 200,000,000	**Date of last Funding** State of last Funding	**27.04.2022**
Management Team	Armin Brun, Alois Schärli, Mark Haller, Marcel Oertle, Andreas Schafer				
Board Members	Antoinette Hunziker-Ebneter, Christoph Lengwiler, Stefan Bichsel, Gilles Froté, Reto Heiz, Annelis Lüscher Hämmerli, Hugo Schürmann, Pascal Sieber, Danielle Villiger				
Key Partners	aity AG (IT), Daura (Tokenisation), myky (exosystem)				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Bitcoin Capital AG

https://bitcoincapital.com/

Bitcoin Capital AG is a Swiss-based subsidiary of FiCAS AG and is focused on issuing Crypto ETPs. The company was founded in 2019 and has listed the world's first actively managed crypto ETP on the SIX Swiss Exchange in 2020. Bitcoin Capital is one of the spearheading crypto ETP issuers and offers innovative and medium to long-term crypto investment vehicles.

Domicile	ZG	**Year of Inception**	2020	**Valuation**	Not disclosed
Employees … of which in CH	**6** 6	**Total Funding** … in 2021	**CHF 100,000** CHF 0	**Date of last Funding** State of last Funding	**03.03.2020** n/a
Management Team	Marcel Niederberger, Roger Suter, Olga Vögeli, Darko Novakovic				
Board Members	Ali Mizani Oskui, Christine Schmid, Mattia Luigi Rattaggi, Luca Schenk				
Key Partners	FiCAS AG, SIX Swiss Exchange, Bitstamp, Crypto Valley Associations, Crypto Compare, Crypto Broker, FTX, coinbase, Kraken, Sygnum, Adexas, incore, Maerki Baumann & Co. AG, Bitcoin Suiss AG, SEBA, Caiac Fund Management AG				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Bitcoin Suisse AG

https://www.bitcoinsuisse.com

Founded in 2013, Bitcoin Suisse Ltd is the Swiss crypto-finance and technology pioneer and market leader. As an enabler for the crypto and blockchain ecosystem in Switzerland, Bitcoin Suisse has been a driving force in the development of the "Crypto Valley" and the "Crypto Nation Switzerland". The crypto-financial services provider offers brokerage, custody, lending, staking, payment solutions and other crypto-related services for private and institutional clients. As a member of the self-regulatory organization Financial Services Standards Association (VQF), Bitcoin Suisse is a financial intermediary subject to Swiss AML/CFT regulations. Bitcoin Suisse consists of several companies under the parent company BTCS Holding Ltd. The company is headquartered in Zug and has built a team of over 300 highly qualified experts in Switzerland and Europe.

Domicile	ZG	**Year of Inception**	2013	**Valuation**	
Employees … of which in CH	**300+** 265	**Total Funding** … in 2021	**CHF 45,000,000**	**Date of last Funding** State of last Funding	**24.07.2020** Series A
Management Team	Dirk Klee, Andrej Majcen, Lothar Cerjak, Markus Perdrizat, Mauro Casellini, Peter Camenzind, Philipp Vonmoos, Ricardo Schlatter, Sven Ramspott				
Board Members	Luzius Meisser, Roger Studer, Arthur Vayloyan, Urs Bigger, Giles Keating				
Key Partners	Worldline, CoinRoutes				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
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BX Swiss AG

https://www.bxswiss.com

BX Swiss AG is an established stock exchange based in Zurich. BX Swiss AG is part of the international Stuttgart Stock Exchange Group and is subject to the Financial Market Infrastructure Act and supervision by the Swiss Financial Market Supervisory Authority FINMA.

Domicile	Zurich (headquarters) & Berne	**Year of Inception**	1884	**Valuation**	
Employees … of which in CH	**16** 16	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Lucas Bruggeman, Matthias Müller, David Kunz				
Board Members	Matthias Völkel, Peter Heller, Dragan Radanovic, Claudio Studer				
Key Partners					

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Covario AG

www.covar.io

Covario is an independent Crypto Prime Broker for institutional clients, bringing together the core components of the global institutional crypto market in a secure, compliant, and scalable platform that covers trading, custody, financing, staking and tailor-made solutions. Covario's platform and services are designed to meet the needs of professional clients with our cutting-edge proprietary technology and experienced team. We connect to all the major trading venues and offer a choice of highly secure custody and financing solutions, together with high-end risk mgmt and reporting tools.

Domicile	ZG	**Year of Inception**	2019	**Valuation**	
Employees … of which in CH	**35** 33	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Mark Banner (CEO), Keith Noyes (CRO & CCO), Patrik Gilli (CFO), Richard Forss (CTO), Eamon Comerford (COO)				
Board Members	Mark Banner (Chair), Maximilian Büttiker, Deepak Gulati, Felix Schmidheiny				
Key Partners	CEXs: 16 exchanges incl. Binance, Coinbase, Gemini DEXs: Curve, SUSHISWAP, UNISWAP Pancakeswap, balancer Custody partners: Fireblocks, Metaco, Copper, Fidelity and Zodia Liquidity providers: Amber, B2C2, Cumberland, DV Chain, Enigma, Galaxy, Genensis, Heh				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Crowdlitoken AG

https://crowdlitoken.com/

First European provider of a digital financial investment for real estate.

Domicile	TG	**Year of Inception**	2018	**Valuation**	
Employees … of which in CH	**5** 4	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	**01.02.2022**
Management Team	Michael Albin Escher, Simon Büchi, Markus Weber				
Board Members	Ernst Sutter, Michael Escher				
Key Partners					

Customer Segments			**Customer Segments by Region**	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			**Key Activities**		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Crypto Consulting AG

https://cryptoconsultingag.ch/

Distribution of an actively managed Crypto Fund which is based on fundamental research. The Fund was launched in 2018 and Assets under Management are at USD 100 million.

Domicile	ZH	**Year of Inception**	2018	**Valuation**	
Employees … of which in CH	**2** 2	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Reto Stiffler, Désirée Velleuer				
Board Members	Reto Stiffler, Désirée Velleuer				
Key Partners	Fund is set up with Sygnum (custodian), Crypto Finance and Bitcoin Suisse and trades on FTX and Binance.				

Customer Segments			**Customer Segments by Region**	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			**Key Activities**		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Crypto Finance Group

https://www.cryptofinance.ch/

CRYPTO FINANCE
DEUTSCHE BÖRSE GROUP

Crypto Finance – a FINMA regulated financial institution and a member of Deutsche Börse Group, offers professional digital asset solutions. This includes one of the first FINMA-approved investment firms with 24/7 brokerage services, custody, infrastructure, and tokenisation solutions for financial institutions, as well as the first FINMA-approved manager of collective assets for crypto assets, with an active, rules-based, and index-based alternative investment fund offering, including the first Swiss crypto fund. Since its founding in 2017, the company has received numerous awards, including Crypto Valley Top 50 Blockchain Company, Top 100 Swiss Start-up, and winner of the Swiss FinTech Award 2019.

Domicile	ZG	**Year of Inception**	2017	**Valuation**	
Employees … of which in CH	**76** 73	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	**01.12.2021**
Management Team	Jan Brzezek, Stijn Vander Straeten, Urs Lehmann, Sarina Christner, Nathaniel Zollinger, Daniel Dettwiler, Patrick Heusser, Bernadette Leuzinger, Lewin Boehnke				
Board Members	Jan Brzezek, Eric Leupold, Alexander Vogel, Hans-Peter Wyss, Uwe Schweickert, Raymond J. Bär, Philippe Cottier, Fabian Schär				
Key Partners	Crypto Finance (Asset Management) AG, Crypto Finance (Brokerage) AG, Crypto Finance (Infrastructure Services) AG				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

daura ag

https://www.daura.ch/

daura

daura is the market leader for the emission of digital participation instruments (shares, participation certificates, convertible bonds) in unlisted companies in Switzerland. daura uses blockchain technology and provides all functions via a user-friendly and legally secure platform. Furthermore, daura enables a seamless integration of digital assets into the existing financial market infrastructure in Switzerland (marketplaces and banks).

Domicile	ZH	**Year of Inception**	2018	**Valuation**	
Employees … of which in CH	**10** 10	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Peter Schnürer (CEO), Claudio Tognella (Sales), Markus Geissler (Product Development)				
Board Members	Armin Brun, Christian Schüpbach, Guido Hüppin, Andreas Kurt Rudolf, Valerio Roncone				
Key Partners	SME-X/Berner Kantonalbank, SIX/SDX, Raiffeisen, BDO, MME, Wenger & Vieli, BV4				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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FICAS AG

https://ficas.com/

FiCAS AG is a Swiss-based crypto investment mgmt boutique. The firm manages FiCAS Active Crypto ETP – the world's first actively managed exchange-traded product (ETP) with cryptocurrencies as underlying assets, issued by Bitcoin Capital AG and offers the issuance of actively managed white-label certificates (AMC).

Domicile	ZG	**Year of Inception**	2019	**Valuation**	
Employees … of which in CH	**7** 7	**Total Funding** … in 2021	**CHF 1,400,000**	**Date of last Funding** State of last Funding	**24.11.2020**
Management Team	Ali Mizani Oskui, Olga Vögeli, Marcel Niederberger, Darko Novakovic, Roger Suter				
Board Members	Sanjeev Karkhanis, Daniel Leo Diemers, Mattia Luigi Rattaggi, Ali Mizani Oskui				
Key Partners	SIX Swiss Exchange, CAIAC Fund Management AG, CryptoCompare, Flow Traders, Synum, coinbase, Bitstamp, Crypto Valley, Kraken, Crypto Broker, incore, Maerki Baumann & Co. AG, SEBA, GrantThornton, Bitcoin Capital AG				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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flov technologies AG

https://www.flovtec.com/

flov technologies AG ("flovtec") is one of world's leading market-makers, connected to more than 120+ exchanges (CEXs & DEXs), and we provide market-making solutions both for liquid & illiquid tokens. Our partners & clients are exchanges, token issuers, protocols, foundations & crypto VCs. We support tokens through-out their whole lifecycle: exchange listings, market-making and smart token execution.

Domicile	ZG	**Year of Inception**	2018	**Valuation**	CHF 18,000,000
Employees … of which in CH	**15** 4	**Total Funding** … in 2021	**CHF 4,700,000** CHF 1,700,000	**Date of last Funding** State of last Funding	**01.11.2021** Seed
Management Team	Khaled Yassin Abdelwahab, Nicolas Grawe, Anton Golub				
Board Members	Frank Klaus Floessel, Florian Wimmer, Daniel Leo Diemers				
Key Partners	Copper, BCB, GenTwo				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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FQX AG
https://fqx.ch/

FQX builds the decentralised debt infrastructure for Web 3.0 based on eNotes™.

Domicile	ZH	Year of Inception	2019	Valuation	
Employees … of which in CH	**18** 9	**Total Funding** … in 2021	**USD 6,700,000** USD 4,7000,000	**Date of last Funding** State of last Funding	**01.06.2021** Seed
Management Team	Benedikt Schuppli, Stephan D. Meyer, Daniel Killenberger, Benjamin Treves				
Board Members	James Courtenay, Benedikt Schuppli, Stephan D. Meyer, David Blumer				
Key Partners	Instimatch Global, CLST, SDX, CM Equity, SIX Fintech Ventures (Investor), Earlybird VC (Investor)				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

FTX Europe AG
https://www.digitalassets.ag/

Tokenising shares, Broker API, Fractional shares and DeFi solutions on blockchain Solana.

Domicile	SZ	Year of Inception	2020	Valuation	n/a
Employees … of which in CH	**5** 1	**Total Funding** … in 2021	**CHF 100,000** CHF 0	**Date of last Funding** State of last Funding	**01.04.2022** 1,000,000
Management Team	Patrick Gruhn, Robin Matzke				
Board Members	Rhotert Max, Matzke Robin, Bavaud Jürg				
Key Partners	FTX US				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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GenTwo Digital AG

gentwodigital.com

GenTwo Digital pioneered the concept of crypto securitisation, were an individual off-balance sheet, issuance vehicle is established to hold digital assets and to issue conventional securities in the form of structured products/certificates with an ISIN (International Securities Identification Number). This allows any financial intermediary (such as banks, asset managers, family offices etc.) to easily warp their digital asset strategy into a bankable security and book crypto on their and or their clients balance sheets. Though this state-of-the-art setup they can conduct traditional audits and can use their traditional banking infrastructure. Trough the ISIN the floodgates to all large- scale/ institutional investors are opened. Think: bankable investment products instead of managed accounts (example financial product term sheet attached).

Domicile	ZG	Year of Inception	2018	Valuation	n/a
Employees … of which in CH	**50** 50	**Total Funding** … in 2021	**n/a** CHF 0	**Date of last Funding** State of last Funding	**n/a** n/a
Management Team	Patrick Rudolf Loepfe, Marco Bumbacher, Philippe André Nägeli, Ralf Hans Glabischnig, Lucas Ereth				
Board Members	Patrick Rudolf Loepfe, Marco Bumbacher, Philippe André Nägeli, Ralf Hans Glabischnig				
Key Partners					

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

InCore Bank AG

https://www.incorebank.ch

InCore Bank is a one-stop business-to-business transaction bank founded in 2007, providing banks, securities dealers, asset managers and fintech companies first-class transaction banking and outsourcing services. As a Swiss corporation with a banking and securities firm license, InCore Bank offers a comprehensive and modular range of services covering transaction banking for traditional and digital assets, business process outsourcing, tailor-made white-label investment products and other services. This offering enables market participants to reduce complexity and focus on their core business.

Domicile	ZH / Schlieren	Year of Inception	2007	Valuation	
Employees … of which in CH	**47.15** 47.15	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Mark Dambacher, René Hertach, Fabian Bieli, Barbara Schreier Kraus				
Board Members	Peter Haist, Hans G. Syz-Witmer, Christian Wunderlin, Sonja Stirnimann, Amedeo C. Vaccani, Daniel Diemers				
Key Partners	MACD, GenTwo Digital, AlgoTrader, KALEIDO, NPB, Kraken, Crypto Finance, Inacta, Finnova etc.				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Leonteq AG

https://www.leonteq.com/

Leonteq is a Swiss fintech company with a leading marketplace for structured investment solutions. Based on proprietary modern technology, the company offers derivative investment products and services. Leonteq acts as both a direct issuer of its own products and as a partner to other financial institutions. Leonteq further enables life insurance companies and banks to produce capital-efficient, unit-linked pension products with guarantees.
The company has offices and subsidiaries in 13 countries, through which it serves over 50 markets. Leonteq AG is listed on the SIX Swiss Exchange (SIX: LEON).

Domicile	ZH	**Year of Inception**	2007	**Valuation**	CHF 1,312,000,000
Employees … of which in CH	**524** 323	**Total Funding** … in 2021	**CHF 802,100,000**	**Date of last Funding** State of last Funding	**01.01.2018**
Management Team	Alessandro Ricci, Markus Schmid, Lukas Ruflin, Marco Amato, Ingrid Silveri, Manish Patnaik, Reto Quadroni				
Board Members	Philippe Le Baquer, Sylvie Davidson, Philippe Weber, Dominik Schärer, Susana Morgado Gomez Smith, Richard Laxer, Christopher Michael Chambers, Thomas Roland Meier				
Key Partners	White-labelling partners: Raiffeisen, EFG International, Cornèr Bank, Aargauische Kantonalbank, PostFinance, Standard Chartered Bank, Basler Kantonalbank, First Rand, Banque Internationale à Luxembourg, Swissquote				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

Maerki Baumann & Co. AG

https://www.maerki-baumann.ch/de

Maerki Baumann & Co. AG is a family-owned private bank. The services range from investment advice to asset management.

Domicile	ZH	**Year of Inception**	1932	**Valuation**	
Employees … of which in CH	**73** 73	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Stephan A. Zwahlen, Lukas S. Risi, Alexander Ising				
Board Members	Hans G. Syz-Witmer, Carole Schmied-Syz, Bruno Gehrig, Urs Lauffer, Michele Moor, Jörg Zulauf				
Key Partners	InCore Bank AG				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

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Mt Pelerin Group SA

https://www.mtpelerin.com/

Mt Pelerin is a regulated Swiss fintech company active since 2018 in financial services bridging traditional finance with the blockchain world. It provides today cryptocurrency brokerage services to retail and business clients around the world, it develops the non-custodial mobile app Bridge Wallet (40k+ downloads) and provides asset Tokenisation services on its platform Bridge Protocol.

Domicile	NE	**Year of Inception**	2018	**Valuation**	CHF 100,000,000
Employees … of which in CH		**Total Funding** … in 2021	**CHF 2,150,000** CHF 0	**Date of last Funding** State of last Funding	**15.12.2018** Seed
Management Team	Sébastien Olivier Moret, Arnaud Salomon, Yann Gerardi, Sébastien Krafft				
Board Members	Sébastien Olivier Moret, Pierre Maliczak, Arnaud Salomon				
Key Partners	Jarvis Network, RealT, BCN, CMTA, Lenz & Staehelin, Crypto Valley Association, ibani				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

Neon Exchange Aktiengesellschaft

https://nash.io/

Nash is bridging the gap between traditional and decentralised finance so anyone can enjoy the new financial opportunities offered by cryptocurrency, combining cash accounts, DeFi-powered earnings, and crypto trading all in one app. With Nash, normal people can grow their wealth through real crypto services that feel just like the digital banking platforms they know.

Domicile	LI	**Year of Inception**	2018	**Valuation**	
Employees … of which in CH	**31** 2 (Liechtenstein)	**Total Funding** … in 2021	**CHF 35,000,000** CHF 0	**Date of last Funding** State of last Funding	**01.09.2018** ICO (security token offering)
Management Team	Fabian Wahle				
Board Members	Fabian Wahle				
Key Partners					

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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SIX Group AG

https://www.six-group.com/de/home.html

SIX operates the infrastructure for the Swiss and Spanish financial centre. The company provides services relating to securities transactions, the processing of financial information, payment transactions and is building a digital infrastructure.

Domicile	ZH	**Year of Inception**	2008	**Valuation**	
Employees … of which in CH	**3,800** 2,500	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Javier Hernani Burzako, Marion Leslie, Jochen Dürr, Marco Menotti, Jos Dijsselhof, Daniel Schmucki, Christoph Landis, Thomas Zeeb				
Board Members	Herbert Julius Scheidt, Andreas Kollegger, André Marc Helfenstein, David Jimenez-Blanco Carrillo de Albornoz, Romana Garcia Belen, Thomas Peter Wellauer, Jürg Gutzwiler, Jürg Bühlmann, Lorenz Erzherzog von Habsburg-Lothringen, Soren Holm Mo				
Key Partners					

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

SwissOne Capital AG

https://www.swissone.capital/

SwissOne Capital is a niche asset manager with a focus on institutional grade crypto and blockchain investment funds.

Domicile	ZG	**Year of Inception**	2018	**Valuation**	
Employees … of which in CH	**5** 2	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Steffen Heinrich Leo Bassler (CEO), Kenny Hearn (Head of Trading), Hugo van Veen (CIO)				
Board Members	Cornelis Jan Quirijns, Antony Turner, Hugo van Veen, Steffen Heinrich Leo Bassler				
Key Partners	Crypto Finance AG, Maerki Baumann AG, APEX Fund Services				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

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Swissquote Bank AG

https://de.swissquote.com/

Swissquote

Swissquote is Switzerland's market leader in online banking. Over three million financial products can be traded on its innovative platforms. Swissquote's core competencies include global stock market trading, trading and custody crypto assets, Forex trading and a Robo-Advisor solution. Swissquote is also active in the payment card, mortgage and leasing markets. As at the end of December 2021, Swissquote held over 55 billion Swiss francs in assets for more than 480'000 private and institutional clients. In addition to its headquarters in Gland, Switzerland, Swissquote has offices in Zurich, London, Luxembourg, Malta, Bucharest, Dubai, Singapore, and Hong Kong. Swissquote holds banking licences both in Switzerland (FINMA) and Luxembourg (CSSF). The parent company, Swissquote Group Holding Ltd, is listed on the SIX Swiss Exchange (symbol: SQN). The Swissquote Group and PostFinance each own 50 percent of the fintech app Yuh Ltd.

Domicile	Gland and ZH	**Year of Inception**	1999	**Valuation**	CHF 3,000,000,000
Employees … of which in CH	**878**	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Marc Bürki, Jan De Schepper, Yvan Cardenas, Morgan Lavanchy, Lino Finini, Alexandru Craciun, Gilles Chantrier				
Board Members	Michael Ploog, Markus Dennler, Monica Dell'Anna, Beat Oberlin, Jean-Christophe Pernollet, Paulo Buzzi, Demetra Kalogerou				
Key Partners	Yuh AG (Joint Venture with PostFinance)				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

Sygnum Bank AG

https://www.sygnum.com/

SYGNUM

Sygnum is the world's first regulated digital asset bank, founded on Swiss and Singapore heritage, operating globally. We make digital assets bankable, secure, and convenient, empowering clients to invest in the digital asset economy with complete trust.

Domicile	Zurich, Switzerland & Singapore	**Year of Inception**	2018	**Valuation**	CHF 783,000,000
Employees … of which in CH	**200** 165	**Total Funding** … in 2021	**USD >170,000,000**	**Date of last Funding** State of last Funding	**01.01.2022** Series B
Management Team	Mathias Imbach, Gino Wirthensohn, Fritz Werner Jost, Martin Burgherr, Martin Jost, Philipp Joel Imbach, Thomas Martin Eichenberger, Fabian Dori, Guido Hüppin, Carolina Thomaz				
Board Members	Peter Wuffli, Gabriela Maria Payer, Kim Leng Chua, Thomas Buess, Luka Müller-Studer, Manuel Caspar Krieger				
Key Partners	Swisscom, Custodigit (joint venture with Swisscom and SIX), Daura (joint venture with Swisscom and SIX)				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
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VanEck
www.vaneck.com

VanEck is a privately held global asset mgmt firm founded in 1955. Recognised for being a pioneer in global markets and for drawing on its experience to offer forward-looking, intelligently designed strategies that seek to strengthen a long-term investment portfolio, VanEck offers strategies to individual investors and institutions, including endowments, foundations, pension plans and private banks across a range of vehicles including mutual funds (retail and institutional shares), pooled institutional funds, variable insurance portfolios, sub-advisory portfolios, separate accounts, exchange traded funds ("ETFs") and UCITS funds.

Domicile	Global	**Year of Inception**	1955	**Valuation**	
Employees … of which in CH	**348** 6	**Total Funding** … in 2021		**Date of last Funding** State of last Funding	
Management Team	Jan van Eck, President, and Chief Executive Officer, Martijn Rozemuller, Chief Executive Officer - Europe Gijs Koning, Chief Operating Officer – Europe, Kristen Capuano, Head of Marketing and Product Strategy John Crimmins, Chief Financial Officer				
Board Members	Gijs Koning, Torsten Hunke, Alexander Baker, Arno Sprenger				
Key Partners	VanEck ETP AG; Arranger, Calculation Agent, and Collateral Agent: VanEck (Europe) GmbH Custodian: Bank Frick & Co AG; Issuing & Paying Agent: Quirin Privatbank AG; Authorised Partici				

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
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Vontobel Holding AG
https://www.vontobel.com

Vontobel specialises in wealth mgmt, asset mgmt and investment solutions for private clients and institutional investors.

Domicile	ZH	**Year of Inception**	1924	**Valuation**	CHF 4,413,000,000
Employees … of which in CH	**2,109** 1,350	**Total Funding** … in 2021	**CHF 56,900,000** n/a	**Date of last Funding** State of last Funding	n/a n/a
Management Team	Zeno Staub, Thomas Heinzl, Maria-Antonella Bino, Felix Lenhard				
Board Members	Herbert J. Scheidt, Andreas E.F. Utermann, Michael Halbheer, Stefan Loacker, Björn Wettergren, David Cole, Maja Baumann, Elisabeth Bourqui, Clara C. Streit, Bruno Basler				
Key Partners					

Customer Segments			Customer Segments by Region	
Retail Clients	Private Clients	Corporates	Switzerland	International (incl. Switzerland)
Family Offices	Banks	Other Institutionals		

Product Offering for Crypto Assets			Key Activities		
Direct Investments	Funds	Infrastructure	Issuing / Tokenisation	Asset Mgmt / Wealth Mgmt	Brokerage
Other Derivatives	Structured Products		Exchange Services	Custody	Lending

References

21Shares. (2021). *Financial Statements.* Retrieved 05/07/2022, from https://cdn.21shares.com/uploads/current-documents/21Shares_AG_2021_IFRS_FS_Mgmt_rpt_%20Audited_FS_final_signed.pdf

Alioth, S., Ankenbrand, T., Bianchi, R., Bieri, D., & Kronenberger, T. (2022, June). *Metaverse Report. An overview of the current status and developments for the financial industry.* Institute of Financial Services Zug IFZ. Retrieved from https://doi.org/10.5281/zenodo.6610900 (+ ID der Publikation: hslu_90865 + Art des Beitrages: Bericht + Sprache: Englisch + Letzte Aktualisierung: 2022-06-03 15:09:22) doi: 10.5281/zenodo.6610900

Ankenbrand, T., & Bieri, D. (2018). *Assessment of cryptocurrencies as an asset class by their characteristics. Investment Management and Financial Innovations*(15, Iss. 3), 169–181.

Ankenbrand, T., Bieri, D., Cortivo, R., Hoehener, J., & Hardjono, T. (2020). *Proposal for a Comprehensive (Crypto) Asset Taxonomy.* In *2020 Crypto Valley Conference on Blockchain Technology (CVCBT)* (pp. 16–26).

Ankenbrand, T., Bieri, D., Frigg, M., Grau, M., & Lötscher, D. (2021). *IFZ FinTech Study 2021. Zug: Lucerne University of Applied Sciences and Arts. Verlag IFZ.*

Ankenbrand, T., Bieri, D., Frigg, M., Kronenberger, T., & Reichmuth, L. (2022). *IFZ FinTech Study 2022.* Retrieved 22/09/2021, from https://blog.hslu.ch/retailbanking/files/2022/03/IFZ-FinTech-Study-2022.pdf

Ankenbrand, T., Bieri, D., Kronenberger, T., Lötscher, D., Sardon, A., Schüpbach, C., & Vincenz, D. (2021). *Crypto Assets Study 2021. Blog HSLU.* Retrieved 2022-05-25, from https://blog.hslu.ch/retailbanking/files/2021/12/IFZ-Crypto-Assets-Study-2021.pdf

Bitcoin Suisse. (2020). *What is DeFi?* Retrieved from https://www.bitcoinsuisse.com/fundamentals/what-is-defi

Bitstamp. (2022). *Bitstamp Crypto Pulse.* Retrieved 2022-05-24, from https://www.bitstamp.net/s/documents/BitstampCryptoPulseReport_Q12022.pdf

Brunner, A. E. (2022). *Swiss Digital Asset Market Report 2022.* Retrieved 30/06/2022, from https://www.homeofblockchain.swiss/report

Coin Metrics. (2022). *Market Cap (USD).* Retrieved from https://charts.coinmetrics.io/network-data/

Coindesk. (2022). *What Is Crypto Custody?* Retrieved from https://www.coindesk.com/learn/what-is-crypto-custody/

CoinGecko. (2022a). *CoinGecko Mainpage.* Retrieved 05/31/2022, from https://www.coingecko.com

CoinGecko. (2022b). *Trust Score Methodology.* Retrieved 08/09/2022, from https://www.coingecko.com/en/methodology

CoinMarketCap. (2022). *Total cryptocurrency market cap.* Retrieved 2022-05-24, from https://coinmarketcap.com/charts/

Crypto Quant. (2022). *Exchange Reserve).* Retrieved from https://cryptoquant.com/asset/btc/chart/exchange-flows/exchange-reserve?exchange=all_exchange&window=DAY&sma=0&ema=0&priceScale=log&metricScale=linear&chartStyle=line

DeFiLlama. (2022). *Total Value Locked (USD).* Retrieved from https://defillama.com/

Dune Analytics. (2022). *Ethereum Beacon Chain Deposits.* Retrieved from https://dune.com/hildobby/ETH2-Deposits

dYdX. (2022). *dYdX V4 - full decentralisation.* Retrieved 08/16/2022, from https://dydx.exchange/blog/v4-full-decentralization

Eurex. (2021). *Digital Asset Trading 2021 - The institutional adoption of a new asset class.* Retrieved 2022-04-06, from https://www.eurex.com/resource/blob/2872736/38c6dd474131994bfe2b45e7e76a0a45/data/20211208_crypto-survey-report.pdf

Gemini. (2022). *2022 Global State of Crypto Report.* Retrieved 2022-05-24, from https://www.gemini.com/gemini-2022-state-of-crypto-global.pdf

Leonteq. (2021). *Annual Report 2021.* Retrieved 05/07/2022, from https://ch.leonteq.com/investor-relations/fyr-2021/leonteq+ag-full+year+report-2021-en.pdf

Lloro, A., Merry, E., Brevoort, K., Jones, K., Larrimore, J., Lockwood, J., … Zabek, M. (2022). *Report on the economic well-being of US households in 2021* (Tech. Rep.). Board of Governors of the Federal Reserve System (US).

Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System.* Retrieved from https://bitcoin.org/bitcoin.pdf

OPSC. (2020). *Bericht finanzielle Lage 2020.* Retrieved from https://www.oak-bv.admin.ch/inhalte/Themen/Erhebung_finanzielle_Lage/2020/Bericht_finanzielle_Lage_2020.pdf

Schär, F. (2021). *Decentralized finance: On blockchain-and smart contract-based financial markets. FRB of St. Louis Review*.

Semrush. (2022). *Dashboard.* Retrieved 05/31/2022, from https://www.semrush.com/projects/

SIX. (2022). *SIX to partner with LMAX Group to launch 24/7 traded crypto futures.* Retrieved 2022-05-18, from https://www.six-group.com/en/newsroom/media-releases/2022/20220317-six-lmax.html

Swiss Post. (online). *Swiss Crypto Stamp Switzerland's first crypto stamp!* Retrieved 2022-07-01, from https://crypto-stamp.post.ch/en

Sygnum. (2021). *Sygnum Bank and Artemundi tokenize a Picasso on the blockchain.* Retrieved 2022-07-01, from https://www.insights.sygnum.com/post/sygnum-bank-and-artemundi-tokenize-a-picasso-on-the-blockchain

Vontobel. (2021). *Annual Report 2021.* Retrieved 05/07/2022, from https://www.vontobel.com/siteassets/about-vontobel/downloads/2022-02-09_fr_full-year-report_en.pdf

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